The official E-magazine of Sabaragamuwa University
Food Science and Technology Association

FOODTECH INSIGHTS

Volume 1 Issue 2 ISSN No. 2950-7065 August 2026

ABOUT FOODTECH-INSIGHT

FoodTech Insights is the biannual e-magazine of Sabaragamuwa University Food Science and Technology Association (SUFSTA).

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A PUBLICATION BY Food Science and Technology Association, Sabaragamuwa University of Sri Lanka

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EDITOR'S NOTE

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Food science is evolving at an unprecedented pace, driven by innovation, sustainability, and the growing demand for healthier and safer food systems. This issue of FoodTech Magazine showcases a diverse collection of articles that explore emerging technologies, novel food products, smart packaging, sustainable protein sources, food safety, digital innovations, and advancements in nutrition and food processing.

From precision fermentation and microbial proteins to digital twins, edible coatings, and food authenticity, the articles featured in this edition highlight how science and technology are shaping the future of food. At the same time, discussions on food laws, ultra-processed foods, media myths, and food safety remind us of the importance of evidence-based knowledge in addressing today's challenges.

I extend my sincere appreciation to all authors, reviewers, and the editorial team for their valuable contributions. I hope this issue inspires curiosity, encourages innovation, and serves as a valuable resource for students, researchers, academics, and industry professionals.

Happy reading!

Dr. Gayani I. Dias
Editor, FoodTech Magazine

Message from the Vice-Chancellor

Message from the Vice-Chancellor

It is with great pleasure that I convey my warm greetings and best wishes to FoodTech Insights, a publication of the Department of Food Science and Technology of the Faculty of Applied Sciences, Sabaragamuwa University of Sri Lanka.

Food science and technology play a vital role in addressing some of the most pressing challenges of our time. As the world faces increasing pressure on food systems due to climate change, population growth, resource constraints, and changing consumption patterns, ensuring food security, food safety, and sustainable food processing has become more important than ever. These challenges call for innovative scientific approaches, technological advancements, and a strong commitment to translating research into practical solutions that benefit society.

Ensuring food security requires not only the availability of adequate food but also equitable access to nutritious, safe, and affordable food. At the same time, maintaining food safety throughout the production, processing, storage, and distribution chain is essential for protecting public health and building consumer confidence. Sustainable food processing technologies, efficient utilisation of raw materials, waste minimisation, and the development of value-added products are equally important in reducing environmental impacts and strengthening the resilience of our food systems.

In this context, technological innovation offers significant opportunities to transform the food industry. Advances in food processing, preservation, packaging, quality assurance, and resource-efficient production can improve product quality, minimise post-harvest losses, and enhance the competitiveness of Sri Lankan food products. Our universities have an important responsibility to foster such innovation through multidisciplinary research, industry collaboration, and the development of graduates equipped to respond to emerging scientific and technological challenges.

I believe that FoodTech Insights can serve as a valuable platform for sharing research findings, disseminating new knowledge, highlighting technological developments, and encouraging dialogue among academics, students, industry professionals, and other stakeholders in the food sector. By bringing scientific insights and practical applications closer together, the magazine can contribute meaningfully to the advancement of food science and technology and to the sustainable development of Sri Lanka's food industry.

I wish to congratulate Dr. Gayani Imanthika Dias, Editor of FoodTech Insights, and all members of the editorial team, academic staff, and students who have contributed to this publication. Their commitment to knowledge sharing and scientific communication is commendable. I encourage them to continue nurturing a culture of inquiry, creativity, responsible innovation, and collaboration that will inspire the next generation of food scientists and technologists.

May FoodTech Insights continue to grow as a respected publication that connects scientific knowledge with societal needs and contributes to a future in which food is safe, accessible, nutritious, sustainably produced, and supported by innovative technologies.

I wish the magazine and its editorial team every success in their future endeavours.

Prof. S. A. Ariadurai
Acting Vice-Chancellor
Sabaragamuwa University of Sri Lanka

Message from the president

Message from the president

As a subject association, we continuously endeavour to disseminate knowledge and emerging trends in food science and technology within the wider community, while actively highlighting the academic excellence of the Degree Programme in Food Science and Technology at the Sabaragamuwa University of Sri Lanka. Continuing our mission of academic outreach and creative dialogue, we are immensely proud to present the latest edition of 'FoodTech Insights'.

This publication represents the collective dedication and intellectual curiosity of many individuals. I wish to convey our deepest gratitude to the Vice Chancellor, the Dean of the Faculty of Applied Sciences, the Head of the Department of Food Science and Technology, and the academic staff members for their steadfast guidance and encouragement. Furthermore, our heartfelt appreciation extends to our alumni, fellow undergraduates, contributing authors, and the tireless editorial committee whose combined efforts made this publication possible. As the President of the Sabaragamuwa University Food Science and Technology Association (SUFSTA), I warmly invite you to explore this issue, and I hope it offers you both inspiration and valuable insights.

About our department

Department of Food Science and Technology

Faculty of Applied Sciences | Sabaragamuwa University of Sri Lanka

Pioneering Excellence in Food Science & Innovation

Rooted in the very inception of higher education in the Sabaragamuwa province, dating back to the university's formative years as the Buttala Affiliated University College (BAUC)—the Department of Food Science and Technology (DFST) has evolved into a premier national hub for food science education, research, and industrial innovation.

Housed within the Faculty of Applied Sciences, the department is committed to producing versatile, ethically grounded, and highly competent food scientists, technologists, and entrepreneurs equipped to meet the evolving challenges of local and global food systems.

Academic & Research Calibre

The four-year Honours Degree Programme is designed to bridge foundational biological and physical sciences with cutting-edge industrial technology. The comprehensive curriculum encompasses:

Core Disciplines: Food Chemistry, Advanced Biochemistry, Food Microbiology, Processing & Preservation Technologies, Food Engineering, and Sensory Evaluation.

Quality & Safety Assurance: Food Quality Management, Industrial Regulations, and Advanced Food Packaging.

Product Development & Enterprise: Food Product Innovation, and Industrial Entrepreneurship.

Backed by laboratories, pilot-scale facilities, and a dedicated academic staff recognized nationally and internationally for scientific excellence, students undergo intensive experiential learning, hands-on practical training, and rigorous independent research projects.

Industrial Synergy & National Recognition

The department maintains robust linkages with leading food processing conglomerates, research institutes, and national bodies such as the Institute of Food Science & Technology Sri Lanka (IFSTSL).

Our undergraduates and staff regularly showcase exceptional talent on national platforms, earning prestigious accolades and

awards at events such as the Pro Food Pro Pack Agbiz Exhibition, Ruhunu Rasa Saraniya, and national food science innovation challenges. The programme concludes with intensive industrial training, ensuring graduates seamlessly transition into leadership roles across quality assurance, research & development, production management, and academia.

The Vibrant SUFSTA Community

Complementing academic rigor is the Sabaragamuwa University Food Science and Technology Association (SUFSTA). Established in 2002, SUFSTA serves as the dynamic heartbeat of undergraduate life. Through community awareness drives, knowledge-sharing platforms, innovation workshops, exhibitions, and publications like 'FoodTech Insights', SUFSTA empowers students to lead, innovate, and contribute meaningfully to society and the global food industry.

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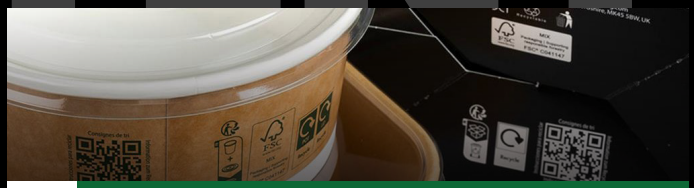

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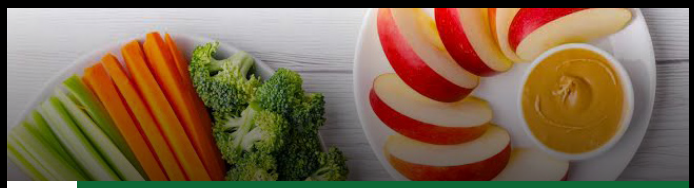

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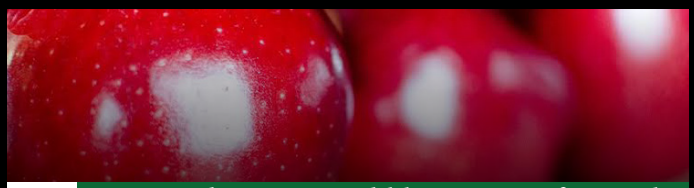

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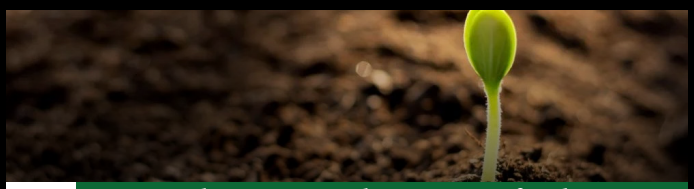

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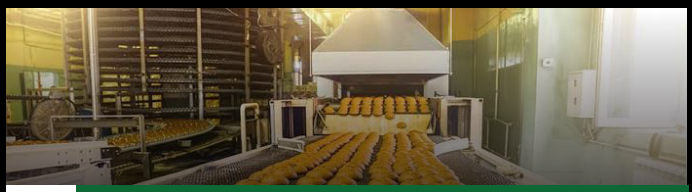

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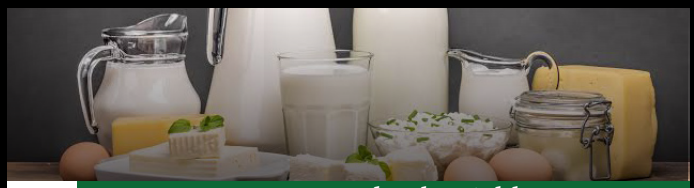

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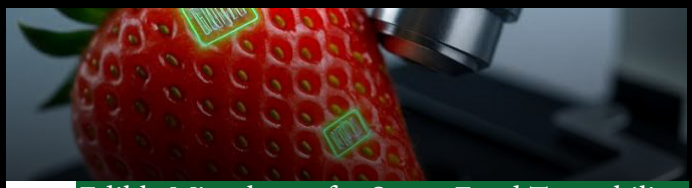

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ISOTOPE RATIO MASS SPECTROMETRY
A SILENT DETECTIVE FOR FOOD AUTHENTICITY

Food fraud has become a global challenge (Laursen et al., 2016). As supply chains lengthen, the food industry needs analytical tools that can see beyond labels and conventional chemical profiles. Isotope Ratio Mass Spectrometry (IRMS) offers exactly that, by reading the subtle "isotopic fingerprints" left by nature and farming practices (Laursen et al., 2016).

Fraud in Real Foods

Organic and conventional crops often differ in $\delta^{15}N$ because manures and composts are typically enriched in 15N compared with synthetic fertilizers (Giannioti et al., 2024). Studies across cereals, vegetables, fruits and animal products show that combining nitrogen with carbon, hydrogen, oxygen and sulfur isotopes can discriminate organic from conventional systems, although legume-based green manures sometimes blur this separation (Giannioti et al., 2024; Laursen et al., 2016).

IRMS is a reference technique for detecting undeclared C4 sugars (from cane or maize) added to C3-based products such as honey and fruit juices because C3 plants have more positive $\delta^{13}C$ values (Laursen et al., 2016; Liu et al., 2023). Similar carbon-isotope strategies are now applied to premium juices and wines to detect sugar addition, water dilution and misdeclared grape origin (Liu et al., 2023).

Natural coconut water from C3 palms shows $\delta^{13}C$ values around -26 to -23‰, whereas common sweeteners from C4 crops lie near -13‰ (Meepho et al., 2021). Elemental analysis - IRMS can detect as little as a 2% addition of single C4 sugars and has shown that roughly half of "no added sugar" commercial products in one Thai survey were in fact adulterated (Meepho et al.,2021).

Stable isotope ratio mass spectrometry has also been widely applied to authenticate animal derived foods, for example to verify beef production systems and to confirm the declared origin

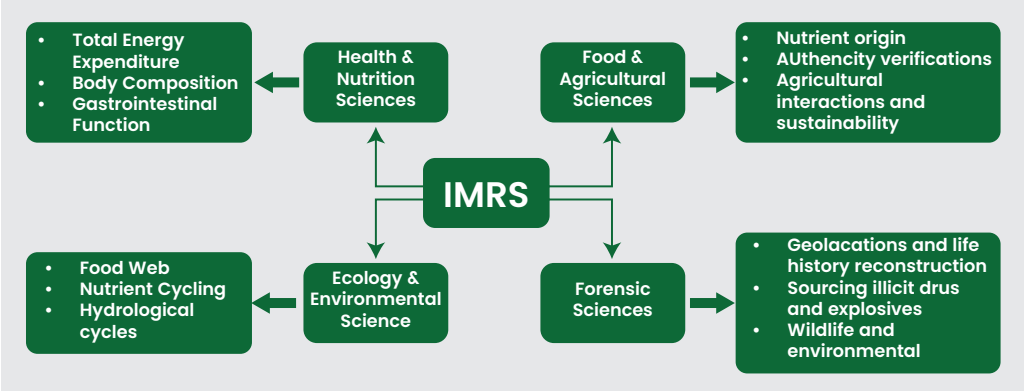

Figure 01 : Applications of IRMS in different fields

of dairy products. Analysis of $\delta^{13}C$, $\delta^{15}N$, $\delta^{2}H$ and $\delta^{34}S$ in bovine muscle and tail hair has been used to retrospectively verify pre slaughter diets, clearly distinguishing cattle finished on pasture from those finished on concentrates or mixed rations, thereby supporting label claims such as "grass fed" or "grain fed" beef (Osorio et al., 2011). Similarly, IRMS of bulk milk and its main fractions (fat, casein and whey) has been combined with chemometric modelling to differentiate Italian milk according to both geographical origin and heat treatment (raw, pasteurised, UHT), providing an effective tool for detecting mislabelled or fraudulently substituted dairy products (Scampicchio et al., 2012).

Compound-Specific Insights

Bulk isotopes sometimes fail when different processes cancel each other out, so attention has shifted to compound-specific isotope analysis (CSIA) where IRMS is hyphenated to gas or liquid chromatography (GC-IRMS or LC-IRMS) (Laursen et al., 2016; Liu et al., 2023). CSIA targets amino acids, fatty acids, individual sugars or aroma.

compounds, providing a much more detailed isotopic picture that is less affected by processing. For example, $\delta^{13}C$ and $\delta^{15}N$ patterns in specific amino acids can separate organic from conventional wheat and rice more reliably than bulk values, while sugar-specific $\delta^{13}C$ greatly improves detection of low-level adulteration in honey, juices and coconut water (Liu et al., 2023).

Limitations of IRMS

In some cases, different production practices can produce very similar bulk isotope values; for instance, the $\delta^{15}N$ value of synthetic fertilizers and green manures can both be close to 0‰, making it difficult to distinguish farming systems based on nitrogen alone (Laursen et al., 2016). Overlapping natural variability in climate, soils and feed, as well as post harvest processing, can also complicate interpretation of bulk isotope data (Liu et al., 2023).

Future Directions

Emerging trends include combining IRMS or CSIA with chemometrics and machine learning to build robust classification models and using laser-based cavity ring down spectroscopy as a faster field screening tool calibrated against IRMS (Liu et al., 2023, Laursen et al., 2016; Giannioti et al., 2024).

Conclusion

IRMS has proven itself as a powerful, science-based tool for combating food fraud. By leveraging the natural isotopic signatures embedded in food products, it provides evidence that is difficult to falsify and complements traditional quality control methods. As technology advances and becomes more accessible, IRMS will play an increasingly vital role in protecting consumers and maintaining the integrity of our global food supply chain.

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FT 02

NATURE'S WARNING SYSTEM

A COLOR-CHANGING FILM FOR SMART PACKAGING

We have all been there. You open the fridge, take out a container of chicken or fish, and pause. It looks fine… maybe. You lean in for the cautious sniff test. Still not sure. The expiry date says tomorrow, but can you trust it?

Meat and seafood are incredibly delicate. Even in the refrigerator, they spoil quickly. As they break down, they release invisible compounds like ammonia and biogenic amines. These are early warning signs of spoilage, but we can't see them. By the time a strong odor develops, the food may already be unsafe. So, we either take the risk, or we throw it away "just in case." That uncertainty leads to food waste, money loss, and sometimes even foodborne illnesses.

What if the container itself could give you a clear answer?

That idea inspired the development of a biodegradable, color-changing film made from something surprisingly simple: corn starch and plant pigments. No electronics. No complicated sensors. Just natural materials working together smartly.

The magic lies in anthocyanins, natural pigments that give plants their beautiful reds, purples, and blues. In acidic conditions, they appear pink or reddish. As the environment becomes more alkaline, they shift toward blue and green. Red cabbage offers a wide color range, while butterfly pea flowers are famous for their deep blue tone. When combined in a balanced 1:1 ratio, they create a stable, highly responsive pigment system, perfect for detecting subtle chemical changes linked to food spoilage.

These natural extracts are incorporated into a corn starch-based film. Corn starch isn't just inexpensive and widely available; it's biodegradable, renewable,

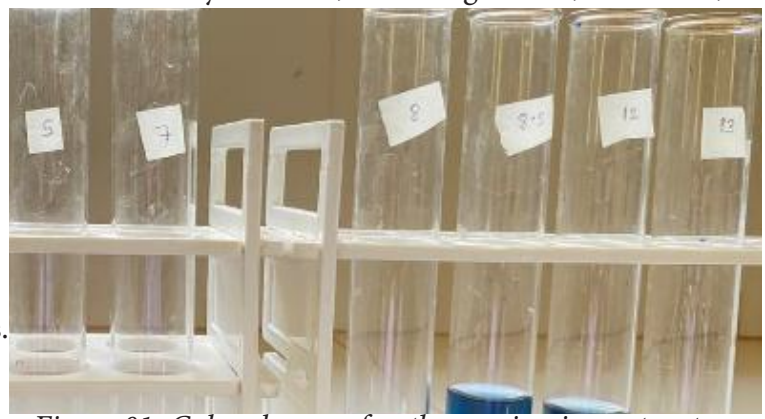

Figure 01: Color change of anthocyanin pigment system with pH change

and capable of forming flexible films. That means the final product doesn't just improve food safety; it also addresses the growing concern over plastic pollution. The film starts with a gentle pink color when the meat or seafood is fresh. As spoilage begins and alkaline compounds accumulate inside the container, the pH around the film increases. Gradually, the color shifts from pink to blue, and eventually green. It's like watching freshness fade in real time. Even better, the film doesn't need to touch the food. It can be fixed inside the lid of a storage container, where it reacts to the gases released during spoilage. This non-contact design keeps things hygienic and practical. You simply glance at the lid, no opening, no smelling, no guessing.

What makes this approach special isn't just science. It's simplicity. A reusable container becomes a smart freshness monitoring system. A natural pigment from flowers and vegetables becomes a silent communicator.

It's a reminder that sometimes innovation doesn't mean adding more technology, it means looking at nature differently. The vibrant purple cabbage and the striking blue butterfly pea flower aren't just beautiful; they're functional. They can help us waste less food, make safer choices, and move toward more sustainable packaging solutions.

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FT 03

ULTRASOUND TECHNOLOGY FOR EFFICIENT FOOD PRESERVATION

Food preservation is an important process that helps ensure the safety, quality, and shelf life of foodstuffs while minimizing postharvesting losses. Foodstuffs are complex materials composed of proteins, carbohydrates, fats, vitamins, minerals, and water, which are prone to physical, chemical, and microbial alterations during storage. For many decades, various preservation techniques, including drying, salting, smoking, refrigeration, and heat treatment, have been employed to prevent spoilage and microbial contamination of foodstuffs (Ercan & Soysal, 2013). Among these techniques, heat treatment is the most employed method due to its efficiency in inactivating microorganisms and deactivating enzymes.

With the increased consumer preference towards consuming less processed, safe, and high-quality food with a natural taste and fewer chemical additives, researchers are now focusing on new non-thermal preservation technologies. One of the new non-thermal preservation technologies is ultrasound technology, which has received considerable attention in food processing in recent decades (Demirdöven & Baysal, 2008).

The ultrasound waves are defined as waves with a frequency higher than 20 kHz, which are not audible to human ears. When ultrasound waves pass through a medium, they cause a change in the medium due to the formation of energy in the medium. This energy causes a phenomenon known as acoustic cavitation, which results in the formation of microscopic bubbles in the medium, causing damage to microbial cells and enzymes (Ravikumar, 2017).

Due to these effects, the use of ultrasound is considered a sustainable preservation method, which is also efficient in improving food quality, reducing

Figure 1: Illustration of ultrasound waves used in food processing. (Source: Adapted from Ravikumar, 2017).

processing time, and minimizing energy consumption. The primary purpose of using ultrasound in food preservation is to extend the shelf life of food products (Altaf et al., 2018).

Microbial Inactivation

One of the most important principles of ultrasound technology is the inactivation of microorganisms. Microorganisms such as bacteria, yeast, and molds are responsible for food spoilage and foodborne illness. The cell membrane of microorganisms is damaged by the application of ultrasound, leading to their inactivation. Sometimes, heat treatment, also known as thermosonication, is applied in combination with ultrasound to inactivate microorganisms.

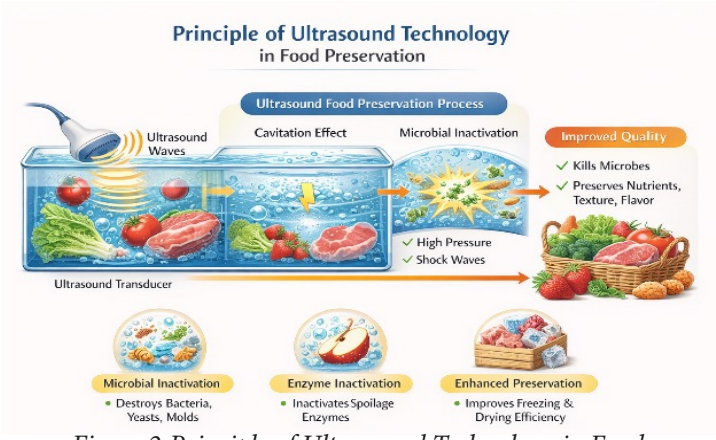

Figure 2:Principle of Ultrasound Technology in Food Preservation. (Source: Adapted from Khadhraoui et al.,2019)

The primary purpose of ultrasound technology is to deactivate enzymes through its application. Enzymes in fruits and vegetables cause browning, softening, and nutrient loss during storage. The application of ultrasound treatment decreases the activity of two enzymes, which include polyphenol oxidase and peroxidase, because they work to preserve food freshness, color and texture (Mañas & Pagán, 2005). Ultrasound in food processing improves preservation by enhancing drying, extraction, filtration, homogenization, and crystallization. It speeds up drying with less energy while maintaining quality and boosts extraction of natural antioxidants and antimicrobials from plants.

Types of Ultrasounds Used in Food Technology

The ultrasound waves employed in food technology can be broadly categorized into low-intensity ultrasound waves and high-intensity ultrasound waves. Low-intensity ultrasound waves are primarily employed in the evaluation of quality and non-destructive testing of food products. On the other hand, high-intensity ultrasound waves generate intense cavitation bubbles and are employed in preservation and processing techniques such as emulsification, freezing, drying, and tenderizing(Khadhraoui et al., 2019).

Limitations and Future Prospects of Ultrasound Technology

Ultrasound technology, although beneficial, still encounters certain limitations in the commercial preservation of food. The intensity, frequency, processing time, and food composition are some factors that affect the efficiency of ultrasound technology. Moreover, the cost of the equipment and its applicability on a commercial scale are significant limitations (Chavan et al., 2022).

Ultrasound technology has achieved recognition today because it serves as a new sustainable technology that benefits the food industry. The system enables energy-saving operations while it enhances food quality and decreases the need for chemical preservatives. The research will confirm ultrasound as an essential tool in modern food preservation and processing.

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LOW-GLYCEMIC INDEX CONFECTIONERY

A NEW APPROACH TO HEALTHIER INDULGENCE

Traditionally, confectionery items are linked to high sugar content and high increases in blood glucose levels. In the backdrop of the rise of lifestyle-related health problems like diabetes, obesity, and metabolic syndrome, the need to develop healthier food options without compromising on the pleasure of consuming food is the need of the hour. In this direction, the low glycemic index confectionery is a new and innovative idea in the food industry, providing the pleasure of consuming confectionery items with healthier metabolic outcomes.

The glycemic index (GI) is a measure used to rank foods containing carbohydrates according to their effect on blood glucose levels following a meal. Foods with high GI values tend to be rapidly digested and absorbed, leading to a significant increase in blood glucose concentration, whereas foods with low GI values tend to release glucose gradually into the bloodstream. Traditionally, confectionery products contain high amounts of refined sugars in the form of sucrose and glucose syrup; thus, they tend to have high GI values. However, modifying these foods to have low GI values is an opportunity as well as a challenge.

One of the major techniques employed in creating

confectionery products with a low GI is the replacement or partial replacement of high-GI sugar with other sweetening alternatives. Polyols such as xylitol, erythritol, and maltitol are commonly used in confectionery products because they have a lower GI and a reduced caloric content. These sweetening alternatives also provide other advantages, such as protection against dental caries, especially in the case of xylitol, which has been shown to prevent bacterial growth. However, overconsumption of these sweeteners can lead to gastrointestinal upsets too, so there must be a limit.

Another significant approach is the use of dietary fibers and complex carbohydrates. Ingredients such as inulin, fructooligosaccharides, and resistant starch have the potential to reduce the GI of confectionery products. These dietary fibers, apart from controlling blood glucose levels, are also prebiotic, providing benefits to the human body. From a food technology viewpoint, one of the major challenges is to ensure that the texture and mouthfeel of confectionery products are maintained.

Protein and fat modification also contribute to the reduction of the glycemic effect of confectionery products. The use of nuts, seeds, milk protein, or plant protein can reduce the rate of glucose absorption by slowing down gastric emptying. The selection of fats or emulsifiers in chocolate-based confectionery products contributes to a slower glycemic effect. The improvement in the functionality of ingredients has enabled manufacturers to balance the nutritional value with sensory attributes.

Low GI confectionery is of particular significance to diabetic patients, weight management, and heart health. Unlike sugar-free products, which are often designed solely to reduce sweetness, low GI confectionery is based on the body's metabolic responses, making it a more "holistic" approach to healthier indulgence. However, it is important to note that regulations are necessary to provide clear labels, avoiding any misleading health claims.

Trends suggest an emerging interest in personalized nutrition and functional confectionery. Low GI chocolates, snack bars, and gummies fortified with fiber or bioactive compounds are emerging as popular products in international markets. In Sri Lanka, where the incidence of diabetes is rising, low GI confectionery products offer opportunities for innovation in the country's food sector, especially in the development of affordable and acceptable products.

Low GI confectionery is a major innovation in the design and use of sweet foods. By applying the principles of food science to the selection and formulation of ingredients, it is possible to develop confectionery products that provide pleasure while facilitating good glycemic control. As consumer knowledge and demand for healthier foods continue to grow, low GI confectionery is likely to be an important part of the future of health-related food innovation.

However, developing low glycemic index confectionery remains challenging, as reducing sugar can significantly affect sweetness, texture, and overall sensory quality. Moreover, the use of alternative ingredients such as polyols and fibers may influence processing stability, shelf life, and consumer tolerance, requiring careful formulation. So as a solution careful combination of different sweeteners and ingredients must be there toimprove the taste and texture. And also, the future research must focus on improving ingredient quality, reducing side effects, and finding affordable local alternatives for wider use.

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FROM PIXEL TO PLATE

PRINTING THE FUTURE OF WHAT WE EAT

How Does 3D Food Printing Work?

Much like conventional 3D printers, 3D food printers deposit material layer by layer based on a digital design file. Instead of using plastic or metal, it uses edible materials called "food inks." These food inks can be obtained from ingredients like chocolate, mashed vegetables, proteins, grains, algae, or plant-based powders. The process starts with a digital design file. This file contains the desired food shape or structure created on a computer. The printer takes this digital plan and deposits the food ink layer by layer building the final edible product. By merging digital design with carefully chosen food materials, 3D food printing enables the creation of customized foods. These foods can have controlled shapes, textures, and nutritional values.

Why Print Food at All?

The value of this technology lies not in replacing traditional cooking but in solving problems it cannot. One of its biggest benefits is nutritional customization. For instance, elderly patients with swallowing difficulties often have to eat pureed foods which are unappealing. This can lead to a decrease in both dietary intake and quality of life. With 3D food printing, these purees can be reshaped into familiar and visually appealing forms, improving both their presentation and acceptance. The technology also allows for precise control over ingredients and nutrients. Meals can be designed to meet the needs of athletes, patients with specific dietary restrictions, or individuals with food allergies. This means that allergens can be completely excluded during formulation and nutrients can be accurately measured. Additionally, 3D food printing can create complex shapes, textures, and layered designs that are hard to achieve manually. Beyond personalization, 3D food printing provides significant opportunities for sustainability and waste reduction. Imperfect fruits and vegetables, food processing by-products, and waste can be turned into usable food inks.

Why Printing Food Is Not as Simple as It Sounds?

Although 3D food printing may seem simple, creating printable food materials is scientifically complex. Food inks need to flow smoothly through printer

nozzles while still holding their shape after printing. Achieving this balance requires careful control of properties like viscosity, elasticity, and structural stability, and it shouldn't compromise the taste, texture, or nutritional value. Temperature control is crucial. Chocolate, for example, must be kept at precise temperatures to maintain its smooth flow and proper crystallization. Also, the printing process itself requires careful calibration. Layer height, extrusion speed, and nozzle temperature must be optimized for each ingredient combination. Too fast, and the layers don't bond properly. Too slow, and the food may overcook or lose structural integrity.

Real-World Applications Today

3D food printing has moved beyond the laboratory and is now being used in various real-world settings. One notable example is space exploration where NASA has investigated this technology for long missions. Through this, meals can be printed from shelf-stable powders rather than relying solely on preserved foods that lose quality over time. This method could give astronauts more nutritious and appealing meals during extended space travel. In healthcare, the technology also lets medications and supplements be incorporated directly into printed foods. This help improve patient acceptance and adherence to treatment. The culinary industry has also embraced 3D food printing. High-end chefs and food designers use the technology to create complex geometric patterns, layered structures, and artistic presentations that would be very difficult or impossible to make by hand.

Why 3D Food Printing Is Not Yet Mainstream?

Despite its potential, several challenges currently limit the widespread adoption of 3D food printing. One major limitation is that it is slow. It can take minutes to print a simple food item and it can take even longer to print something more complex. This makes the process less practical compared to conventional cooking methods. Another challenge is ingredient compatibility. Not all foods can be easily converted into printable food inks. Fresh fruits and vegetables, for example, do not naturally possess the flow and structural properties required for extrusion-based printing. Researchers are still working to develop formulations that maintain both printability and nutritional quality. Cost also remains a significant barrier. Professional-grade 3D food printers and specialized materials are expensive, restricting accessibility for most consumers and small businesses. However, as the technology advances and production scales increase, prices are expected to decrease over time. Regulation presents an additional obstacle. 3D food printing needs to be safe and consistent. Existing food safety laws were not developed with 3D food printing in mind, and internationally standardized guidelines for printed foods are still evolving.

> **"From pixel to plate, we're not just printing food; we're printing possibilities"**

The Next Decade

Looking ahead, several trends suggest how 3D food printing might evolve over the next decade. Just as microwave ovens revolutionized home cooking in the 1970s, compact 3D food printers may become common kitchen appliances.

3D food printing challenges the traditional linear path from farm to fork which is growing, harvesting, processing, packaging, shipping, and finally cooking our food. Instead we might soon design, download, and print our meals. This technology raises concerns about losing the social and cultural aspects of food preparation, the skills passed down through generations. As 3D printing matures, it should complement, not replace, the timeless human practices of preparing and sharing food with the people we care about

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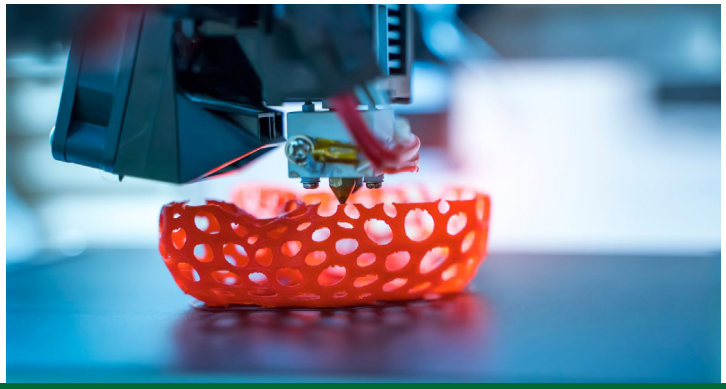

PRECISION FERMENTATION

ENGINEERING MICROBES TO PRODUCE ANIMAL-FREE PROTEINS

The global food system will have to deliver far more protein by 2050 within environmental limits, yet conventional livestock farming creates around 14-15% of global greenhouse gas emissions, uses a large amount of land, and puts heavy stress on water resources (Poore and Nemecek, 2018; Mads Pagh Nielsen et al., 2023). It is against this background that precision fermentation has emerged as a biotechnology that programs microorganisms to produce animal proteins without raising animals (Nielsen et al., 2023).

Precision fermentation employs recombinant DNA methods, unlike the traditional fermentation used in making yogurt or beer (Mads Pagh Nielsen et al., 2023). In precision fermentation, genes that encode specific animal proteins such as whey, casein, egg proteins, and collagen are introduced into microbes such as bacteria, yeasts, fungi, and algae (Mads Pagh Nielsen et al., 2023; Liu et al, 2024). The resulting engineered cells synthesize proteins that can be identical in structure or very similar to their animal-derived counterparts, although protein functionality is influenced not only by amino acid sequence but also by folding and post-translational modifications (Mads Pagh Nielsen et al., 2023). The proteins are generated during normal cellular metabolism, then harvested, purified, and incorporated into foods; however, downstream processing can be complex and expensive (Knychala et al., 2024). Since the amino acid sequence and overall structure are designed to match the originals, these ingredients are intended to provide comparable functionality and nutritional value, although exact performance depends on the final molecular and processing context (Mads Pagh Nielsen et al., 2023).

Delivery of this process at scale requires advanced strain engineering and process optimization (Knychala et al., 2024; Mads Pagh Nielsen et al., 2023). First, genes are selected from animals, such as dairy proteins from cattle or chicken egg proteins (Mads Pagh Nielsen et al., 2023). These are then introduced into host microorganisms like Saccharomyces cerevisiae, Trichoderma reesei, or Yarrowia lipolytica (Knychala et al., 2024). An appropriate host should be capable of achieving a very high protein titer, usually above 50 g/L, and secreting proteins effectively into the culture medium; it should grow quickly on low-cost carbon sources such as glucose or glycerol and fit into food-grade fermentation systems (Mads Pagh Nielsen et al., 2023). In large fermenters, tem-

perature, pH, dissolved oxygen, and nutrient levels are kept within tight ranges to maximize yield and maintain quality (Knychala et al., 2024).

Commercialization is rapidly moving from pilot to market (Mads Pagh Nielsen et al., 2023; Rice et al., 2025). Companies like Perfect Day are launching precision-fermented dairy proteins, using recombinant whey and casein in products like ice cream, yogurt, and protein powders that aim to deliver conventional dairy performance without cows (Mads Pagh Nielsen et al., 2023). Other companies are working on precision-fermented collagen, gelatin, and heme to improve the texture, juiciness, and overall sensory properties of plant-based meat and dairy alternatives (Rice et al., 2025; Kirtil, 2025). Early products generally position these proteins as specialty ingredients that enhance mouthfeel, texture, and nutritional completeness (Kirtil, 2025; Rice et al., 2025).

One of the main reasons precision fermentation has attracted significant interest is its potential environmental benefit (Liu et al, 2024; Knychala et al., 2024). Studies have suggested that precision fermentation could reduce land use by around 90-95% compared with conventional dairy production, while also reducing water use and feed demand (Knychala et al., 2024). It may also reduce greenhouse gas emissions by approximately 75-95% compared with cattle-based production systems (Knychala et al., 2024). Moreover, antibiotic use can be minimized because the process does not depend on rearing animals (Liu et al, 2024; Knychala et al., 2024).

However, several challenges limit the full commercialization of precision-fermented proteins (Knychala et al., 2024; Rice et al., 2025). For instance, they are still relatively expensive when compared to animal-derived proteins; therefore, achieving cost equivalence requires production at very large scale and lower cost (Knychala et al., 2024; Mads Pagh Nielsen et al., 2023). The time required for regulatory approval also varies, with some countries moving faster than others and Europe generally taking a more cautious approach (Rice et al., 2025). Food perception issues also remain a challenge, particularly around foods associated with genetic engineering and unfamiliar production methods (Grundy et al., 2024; Rice et al., 2025).

Future work will continue to focus on two major areas: economics and eating quality (Kirtil, 2025; Rice et al., 2025). Producing high-quality sensory experiences will depend on carefully integrating precision-fermented proteins with plant-based systems, fats, flavors, and new texturization techniques (Kirtil, 2025). As strain engineering and bioprocess technologies continue to improve, precision fermentation is likely to expand beyond high-end markets, integrate with cultivated meat and plant-based systems, and become an important part of a more sustainable and climate-friendly protein sector (Rice et al., 2025).

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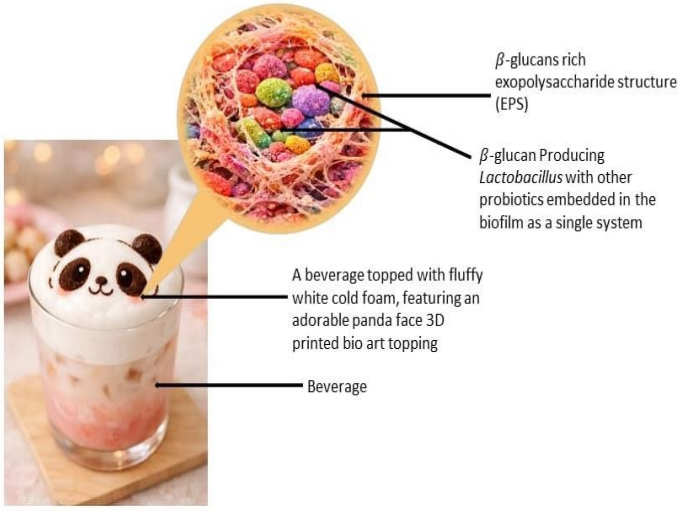
EDIBLE LIVING MICROBIAL TOPPINGS

A NOVEL APPROACH TO PROBIOTIC FUNCTIONAL FOODS

As I sipped my latte, my eyes followed the intricate swirls across its surface. In that quiet moment, a question arose: What if food art could come to life? Could we integrate microbes into this creative space, reimagining functional foods that are both visually captivating and biologically active?

The illustration shows a beverage topped with a functional foam featuring 3D-printed bio art with panda face. The magnified picture shows the microscopic structure consisting of a β-glucan-rich exopolysaccharides matrix embedded with Lactobacillus with other probiotics, forming stable, edible biofilm system.



Functional foods are defined as those that provide benefits beyond basic nutrition. They contain bioactive compounds that improve physiological functions, reduce disease risk, or enhance overall well-being (Fekete et al., 2025). Probiotics are live microorganisms that confer measurable health benefits when consumed in adequate amounts which play a central role (Sanders et al., 2018). They are often incorporated into dairy products like kefir and yogurt, but have now expanded into non-dairy formulations, including juices, sparkling sodas, and functional waters (Anand and Awasti, 2020). Foods that incorporate probiotics are commonly referred to as probiotic functional foods. Although these foods offer substantial potential for enhancing nutritional quality and health benefits, probiotics face significant challenges during food processing and storage, particularly in terms of their survival, stability, and shelf life.

In addressing the unanswered questions regarding the stability and survivability of probiotic functional foods, bio-art toppings may provide novel insights. Traditional artificial food toppings often have viscous, gel-like, or elastic textures that maintain their structure when decorating food. Surprisingly, these

viscoelastic textures, termed ropy matrices, are naturally produced by β-glucan-synthesizing Lactobacillus strains, many of which are recognized probiotics, due to the extracellular secretion of high-molecular-weight β-glucans that impart increased viscosity and elasticity (Zhang et al., 2023).

These Lactobacillus strains can tightly link to proteins and effectively fill the network structures of the β-glucan matrix, resulting in high viscosity characteristics. Therefore, utilizing β-glucan producing *Lactobacillus* strains for bio-art toppings holds considerable technology for improving gel quality. In addition, β-glucans improve medium viscosity and rheological stability through their specific branching structures and the formation of a heat-induced, three-dimensional gel network. This contributes to thicker textures and resilience against both acid and heat stresses (Stack et al., 2010). β-glucans are also associated with numerous health-promoting and prebiotic properties, including anticancer, anti-inflammatory, and immunomodulatory effects, and they enhance the adhesion of beneficial microbes to human intestinal epithelial cells, thereby supporting probiotic survivability (Russo et al., 2012). β-glucan-producing *Lactiplantibacillus plantarum* (*Lactobacillus plantarum*) strains. These strains form dense, ropy matrices that increase gel viscosity and water-holding capacity. This capability not only improves the texture of beverages but, when co-cultured with other probiotic microorganisms, they create a cohesive and stable environment by generating a protective β-glucan matrix (Hernández-Figueroa, López-Malo and Mani-López, 2025). Therefore, artificial toppings could potentially be replaced by *Lactobacilli strains*.

To engineer microbial consortia that produce β-glucans and beneficial probiotic properties, a biofilm mode of these microorganisms may be ideal. In biofilm mode, microbes come together to form communities protected by a shield of polysaccharides, proteins, glycoproteins, and glycolipids. Notably, *Lactobacill*i strains, particularly *Lactiplantibacillus plantarum* (*Lactobacillus plantarum*), also form the biofilms with an extracellular matrix with β-glucans that enhance gastrointestinal resistance and probiotic functionality. (Bugday et al., 2024). Therefore, if combining such β-glucans producing *Lactobacilli* strains with beneficial probiotics, these biofilms may serve as stable and functional addition. Furthermore, these structures can integrate into 3D printed toppings by utilizing 3D printing technology. Paste-like food materials used as inks in extrusion-based 3D food printing, showing that water content strongly controls yield stress and elastic properties while preserving overall flow behavior in multi-component systems. (Wei et al., 2023). To achieve this, probiotic biofilms enriched with the β-glucan-rich *Lactobacilli* strains could be formulated as they are naturally having rheological and water holding capacity. In such 3D-printable probiotic toppings both structural integrity and probiotic functionality will preserve. Additionally, to enhance consumer engagement and aesthetics, natural pigments or bioactive dyes can be incorporated into this β-glucan rich probiotic biofilm matrix. This will result in vibrant, multi-colored bio toppings on beverages transforming ordinary beverages into high-potency, visually engaging probiotic delivery systems.

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THE COFFEE PARADOX

FROM COMFORT TO JITTERS

Coffee is the world's most widely used source of caffeine, which is a psychoactive substance that is relied upon for its reliable jolt of alertness. But each individual's experience varies vividly. While one person finds calm and focus, another may feel jittery or anxious. This duality reflects the scientific paradox in coffee's chemistry and humans' genetic blueprints. Modern research shows that coffee sets off a complex chain of reactions in the brain.

Caffeine's action in coffee is responsible for its stimulant and potential anxiety-inducing effects. Caffeine acts as an antagonist in adenosine receptors. Adenosine is a neuromodulator which promote sleepiness and reduces neural activity by binding to adenosine receptors (primarily the A_1 and A_2A subtypes). Under normal conditions, adenosine accumulates in the brain throughout the day, causing fatigue and encouraging rest. Caffeine's molecular structure can block these adenosine receptors and can prevent the action of adenosine from signaling fatigue. As a result, the brain does not receive the signal to slow down, leading to increased alertness and wakefulness.

The action of caffeine triggers a cascade effect. The blockade of adenosine receptors causes a strong effect on dopamine, the neurotransmitter which responsible for pleasure, attention, and regulating the brain's reward system, and norepinephrine, which is responsible for the "fight-or-flight" response, alertness, and cognitive function. However, in susceptible individuals or at high doses this actions are getting mimicked and create a stress

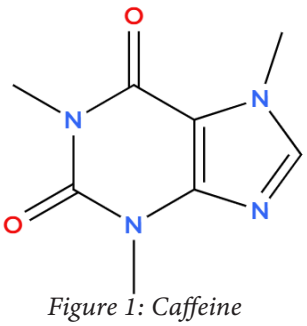

Figure 1: Caffeine structure

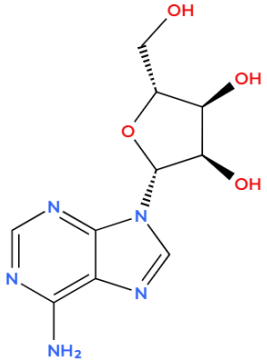

Figure 2: Adenosine structure

response and produce anxiety, which is called the anxiogenic (anxiety-provoking) pathway of caffeine.

People with certain genetic variations in the ADORA2A gene (Gene responsible for the formation of A_2A adenosine receptor) report higher anxiety after caffeine intake. Even moderate-level caffeine content can stimulate the sympathetic fight responses, such as increasing heart rate, sweating, and restlessness, in these individuals. However, this anxiety effect is dose and habit-dependent. Habitual drinkers of coffee tend to develop more tolerance, while naïve consumers may be more sensitive.

Brain Boost or Brain Buzz?

- Over ***2.25 billion cups of coffee are consumed every day worldwide***, making caffeine the most widely used brain-active substance on Earth.
- Caffeine indirectly enhances dopamine signaling linked to motivation and focus, but its effect is much milder than drugs like nicotine or cocaine.
- After drinking coffee, caffeine is absorbed into the bloodstream within about 30–45 minutes and easily crosses the blood–brain barrier.
- Up to ***400 mg of caffeine per day*** (about 3–4 cups of coffee) is considered safe for healthy adults (EFSA, 2015).

Coffee is not all about caffeine. While caffeine delivers the acute stimulant effect of coffee, the other bioactive compounds may be the key to its soothing and protective effects. Coffee is a complex mixture of over 1000 phytochemicals. Polyphenols such as chlorogenic acids (CGAs), caffeic acid, and trigonelline are the major ones among these. These compounds possess antioxidant and anti-inflammatory properties that support brain health.

As epidemiological studies suggest, regular consumption of coffee lowers the risks of neurodegenerative diseases such as Parkinson's and Alzheimer's disease. Moreover, it reduces the incidence of depression and cognitive decline. These associations are the result of the combined effect of multiple coffee compounds. For instance, CGAs may cross the blood-brain barrier and can reduce neuroinflammation

Why Each Experience Is Unique?

The most intriguing factor in the coffee paradox is genetics. CYP1A2 and ADORA2A are the two major genes responsible for this unique experience. 90% of the caffeine is metabolized in the liver, and CYP1A2 determines the speed of this process. In the liver, caffeine is primarily metabolized by the CYP1A2 enzyme into paraxanthine, theobromine, and theophylline. Individuals with certain variations in the gene CYP1A2 can quickly metabolize caffeine in the liver. Clearing caffeine faster has beneficial effects. Slow metabolizers retain caffeine longer, which can amplify the coffee's stimu-

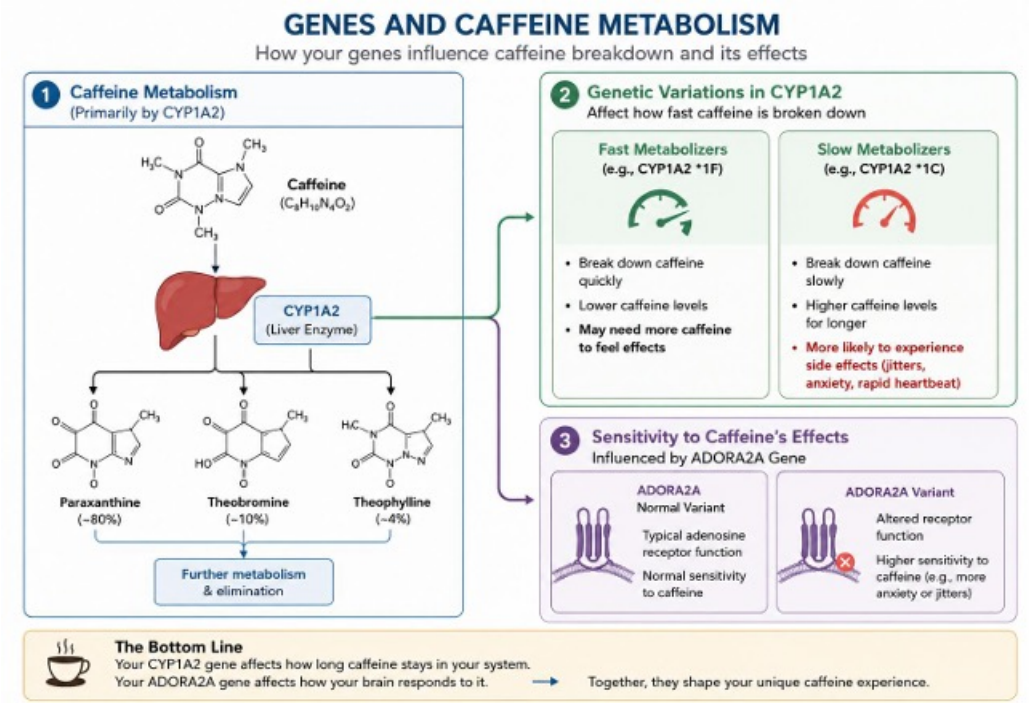

Figure 3: Caffeine metabolism and genetic influence. Caffeine is metabolized in the liver by the CYP1A2 enzyme into major metabolites, while ADORA2A gene variations influence individual sensitivity to its neurological effects.

lant and anxiogenic effects.

The gene ADORA2A is responsible for the coding of the A_2A adenosine receptor, which caffeine blocks. Individuals with variations in this receptor gene are highly sensitive to caffeine-induced anxiety. Even a small amount of caffeine can overactivate their brain's emergency response network.

The paradoxical effect of coffee, which both calms and energizes people, demonstrates how food bioactives interact with human biological systems. A single cup of coffee contains a diverse mixture of compounds that interact with the brain in multiple ways, and individual genetic differences further shape these responses. The intricate relationship between both elements allows us to view coffee as more than a common drink because it serves as a scientifically interesting functional beverage whose effects differ for each individual

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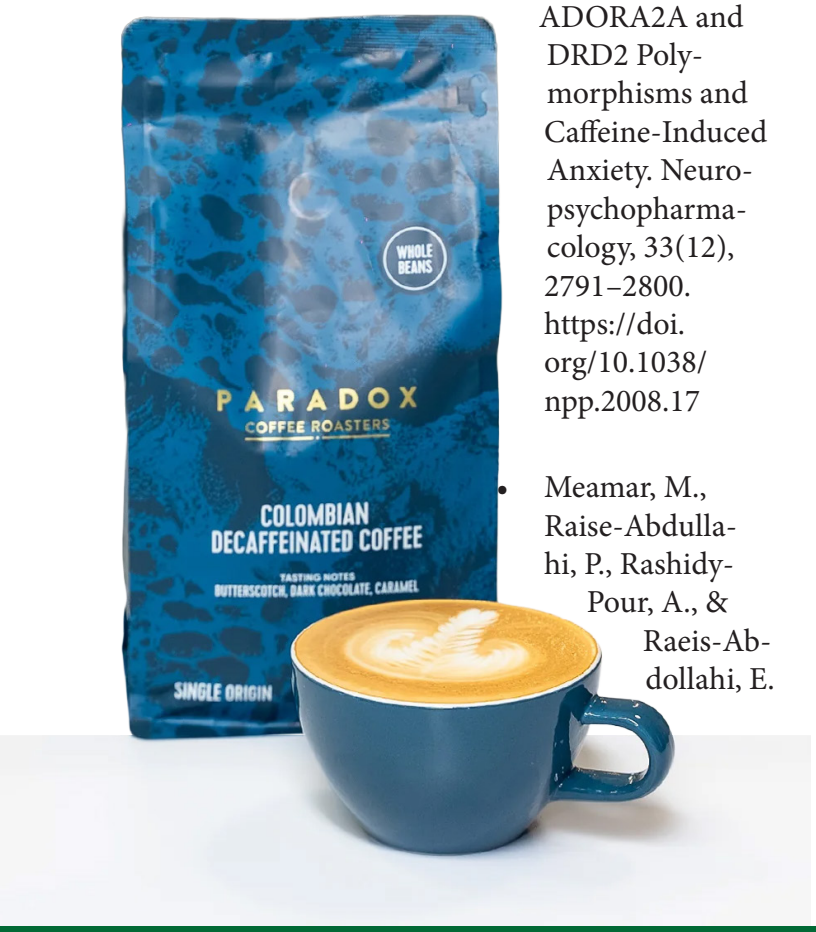

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BEYOND THE NEEDLE

THE FUTURE OF VACCINATION GROWN IN PLANTS

What if your next vaccine came from a plant instead of a syringe? Plant-made vaccines are turning this possibility into reality.

Why has the world acquired plant-based solutions?

Traditional vaccines tend to use live- attenuated viruses or proteins made in cell cultures, along with chemical adjuvants to boost immune responses. Conventional processes are more costly, time-consuming, and pretty much centralized, as access to the vaccines becomes a huge burden for developing countries. The COVID pandemic clearly exposed these weaknesses, such as shortages, limited production, and even cold chain difficulties. As a result, billions of people were left without adequate protection (Rybicki, 2010). Plant-related vaccines are considered promising alternatives.

From fields to pharmacies: The Concept of Plant-Vaccines

In the early 90s, scientists discovered that plants can produce similar proteins to those in animal cells. Therefore, the concept of using plants as bioreactors is becoming more popular. The specific gene encoding an antigen is inserted into plant genetic material using an engineered soil bacterium called Agrobacterium or a gene gun. Then the plant reads the foreign DNA and manufactures the target protein, acting as a bioreactor. After consuming these plant-based products or processed vaccines, antigens can directly activate the immune system and help to fight against harmful pathogens. This method was applied in several conventional crops, mentioned in Table 1 as clinical trials (Takeyama et al., 2015). These plants are becoming green factories with the ability to produce pharmaceuticals. Compared to traditional vaccine production, these plant-made vaccines are a more sustainable alternative. They can be grown, harvested, and distributed with minimal infrastructure. This approach represents a remarkable step toward bringing the future of medicine quite literally from fields to pharmacies

Science behind Edible Therapeutics

Biotechnology is the key path to edible therapeutics production, where they deal with genetically engineered plants. These genetically engineered plants are used as natural bioreactors capable of synthesizing recombinant proteins in a cost-effective and scalable manner. These capabilities can be used to produce

Table 1. Plant-based human vaccines in clinical trials

Plant	Antigen	Pathogen or disease	Administration route
Transgenic potato	Heat-labile toxin B subunit (LTB)	Enterotoxigenic *Escherichia coli*	Oral
	Capsid protein	Norovirus	Oral
	Viral major surface protein	Hepatitis B virus	Oral
Transgenic Rice	CTB	Cholera	Oral
Transgenic maize	LTB	Enterotoxigenic *Escherichia coli*	
Transgenic lettuce	Viral major surface protein	Hepatitis B virus	Oral
Transient spinach	Glycoprotein, nucleoprotein	Rabies virus	Oral
Transient *Nicotiana benthamiana*	HA	Influenza virus	Intramuscular

biologically active therapeutic proteins for disease management and other commercial uses. These edible therapeutics are produced by introducing human therapeutic genes into the plants' genomes, allowing plants to produce functional proteins in edible tissues. These produced proteins are stored in edible plant parts such as seeds, protective plant cell walls, or tubers.

According to He et al. (2011), they inserted the human gene encoding serum albumin production into the rice genome. The gene sequence is read by the transcription and translation machinery in the inserted gene of the plant, and recombinant proteins are synthesized during seed development. These human serum albumins, which are produced by the plant shown to be structurally and functionally equivalent to plasma-derived albumin (He, et al., 2011). According to Daniell et al. (2023), they indicated a more advanced concept of oral delivery without purification of therapeutics in proinsulin-producing genetically engineered lettuce. In this study, the human proinsulin proteins were encapsulated within plant cell walls and protected from degradation in the stomach. When plant materials reach the intestine, gut microbes and digestion release the proteins. Afterwards, proinsulin is converted into active insulin and regulates the blood glucose level. This system works as a natural drug delivery capsule built by the plant itself.

Recent Breakthroughs

Scientists used fast-growing plants like tobacco to produce COVID-19 vaccines. They engineered these tobacco plants to create virus-like particles that mimic the SARS-CoV-2 virus structure (Ward et al., 2021). They have not made anyone sick, as they basically stimulated the body's immune system. This plant-based vaccine technology might be applied beyond COVID-19 to various diseases such as influenza, hepatitis B, norovirus, and Ebola (Kurup & Thomas, 2020). Edible therapeutics take this concept further, applying it to a broader range of therapeutic uses. This method solves major healthcare access issues by allowing people to receive vaccines through food, without medical professionals or complex storage.

Benefits and Challenges

Benefits

- They offer cost benefits and are more environmentally friendly.
- They reduce the need for expensive bioreactors.
- This method allows for quick scaling, which is crucial during a pandemic or sudden breakouts.
- Plant systems are also generally protected from human pathogens and end up safer than some of the usual cell culture vaccines.

Challenges

- Regulatory approval for vaccines made in plants is just getting started in many countries.
- Maintaining consistent dosage levels in edible vaccines is a challenge.
- Public acceptance of genetically modified foods changes depending on the culture.

Researchers solve these challenges with better purification techniques and add rigorous quality control. Public education campaigns help build trust in these new kinds of therapies (Kurup & Thomas, 2020).

Plant-made vaccines and edible therapeutics have the potential to revolutionize global healthcare. This approach makes vaccines easier to obtain, cheaper, and more people-friendly. That means significantly reducing infectious diseases, particularly in places with low income. It might extend to managing chronic conditions, allergies, and cancer treatments through the immune system (Daniell, et al., 2023). All this demonstrates how plant biotech has a huge impact beyond just food.

The merging of farming and medicine offers a sustainable, inclusive solution for global health, potentially revolutionizing prevention and treatment through everyday foods.

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NATURE'S FRESHNESS SHIELD

RECENT ADVANCES IN EDIBLE COATINGS FOR FRESH PRODUCE

Introduction

Fresh fruits and vegetables are perishable, especially due to high moisture content and metabolic activities that happen after harvesting. Significant post-harvest losses globally occur due to respiration, moisture loss, oxidation, and microbial activity in fresh produce, characterized by elevated moisture levels and ongoing metabolic activities (Umeohia & Olapade, 2024). With the current trends of consumer demand for safe, high-quality, and minimally processed foods, edible coatings have been considered a potential technology for the preservation of perishable fresh produce. These act as protective films that cover the surface of fresh fruits and vegetables, thereby acting to extend shelf life (Vidthyaa et al., 2025).

Concept and Mechanism of Edible Coatings

Edible coatings may be defined as the application of very thin films of food-grade materials directly onto the surface of the target product (Armghan Khalid et al., 2022). These are often derived from polysaccharides such as starch, cellulose, alginate, and chitosan, as well as proteins like whey and lipids. The primary function of edible coatings can be characterized as an organic gas exchange medium that regulates gas transport, mitigates moisture loss, decreases respiration rate, and safeguards against microbial proliferation(Jama et al., 2026). Moreover, edible coatings exhibit biodegradability, distinguishing them from other food products.

Recent Technological Advances in Edible Coatings

Recent research focuses on improving edible coating by the addition of natural bioactive compounds such as essential oils (cinnamon, clove), plant extracts, and organic acids to inhibit microorganisms and improve food safety. Nano-enhanced coatings with particles like nano-chitosan, Nano-cellulose, and nano-emulsified essential oils further result in improved uniformity and functional properties (Gidagiri et al., 2025). Also, polysaccharide-protein-lipid composites provide higher mechanical strength and reduced permeability than single-component coatings.

Real-world Applications in Fresh Produce Preservation

The application of edible coatings has proven high-

ly successful for a wide range of fresh fruits and vegetables, including apples, bananas, mangoes, strawberries, citrus fruits, tomatoes, cucumbers, and capsicums. In real-world food supply chains, these coatings are increasingly used during post-harvest storage, long-distance transportation, and retail display to maintain freshness and reduce quality deterioration. By forming a thin protective layer on the surface of produce, edible coatings help minimize moisture loss, delay ripening, preserve firmness, and extend shelf life by several days or even weeks depending on the type of fruit, coating material, and storage conditions(Sanjay et al., 2022). As a result, fruits and vegetables remain attractive, fresh, and marketable for longer periods. This is especially important for exporters, supermarkets, and food distributors, as extended shelf life reduces post-harvest spoilage, lowers economic losses, and increases market value by improving the quality and consumer acceptability of fresh produce. Among different coating materials, chitosan-based coatings have shown remarkable antifungal properties, helping to control post-harvest diseases and significantly reduce microbial spoilage during storage and transport. Overall ,edible coatings contribute greatly to maintaining product quality, texture and moisture content while supporting sustainable and cost-effective food preservation practices(Miteluț et al., 2021).

Challenges and Future Perspectives

Though edible films and coatings possess many benefits, their application is still associated with several challenges. Improper formulation or excessive application of these coatings may negatively affect the sensory qualities of fresh produce, including flavor, aroma, texture, and color, which can reduce consumer acceptability. In addition, differences in coating thickness, storage conditions, and compatibility with different fruits and vegetables may influence their overall effectiveness. However, these challenges can be minimized through proper optimization of coating formulations, controlled application techniques, and the use of natural favor-compatible ingredients that maintain the original sensory properties of the product. Researchers are also focusing on developing advanced smart edible coatings that can respond to environmental changes such as temperature and humidity fluctuations, improving the stability and performance of coated products during storage and transportation. Furthermore, growing interest in the use of natural materials and locally available agricultural processing by-products for edible coating production may help reduce production costs, improve sustainability, and increase consumer acceptance of these eco-friendly preservation technologies.

Conclusion

Edible coating is an innovative, environment-friendly technique for fresh produce preservation and reduce of post-harvest losses. In recent years, active, nano-enhanced, and composite coating technologies have greatly advanced their functionality. Subject to continued research and technological optimization, edible coatings may play a key role in the partial replacement or complementing of conventional packaging systems in the food supply chain of the future(Bremenkamp & Sousa Gallagher, 2025).

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FT 11

FOOD FOR HEALTH AND WELFARE

A COMPARATIVE STUDY OF FOOD LAW IN SRI LANKA AND INDIA

Food regulation is an important legal mechanism used by states to protect public health and ensure the availability of safe and adequate food. In Sri Lanka, food regulation is primarily governed by the Food Act No. 26 of 1980, together with the Food (Amendment) Act No. 20 of 1991, which focus on food safety, prevention of adulteration, labelling, registration, inspection, and prosecution. In India, food regulation operates through a broader legal framework that includes food security and nutrition, mainly under the National Food Security Act, 2013, which provides statutory entitlements to food for identified sections of the population (Patel and Nagar, 2014). This comparative analysis examines how food law is structured and implemented in both jurisdictions by referring to statutory provisions, institutional arrangements, and judicial interpretation.

In Sri Lanka, the primary objective of food law is to ensure that food available for human consumption is safe and free from adulteration. The Food Act No. 26 of 1980 prohibits the manufacture, import, sale, and distribution of food that is injurious to health, unfit for human consumption, or falsely labelled (Fernando, 2007). The Act empowers health authorities to regulate food handling, storage, packaging, and advertising (Karunasena, 2010). In contrast, India's National Food Security Act, 2013 goes beyond food safety and focuses on ensuring access to adequate food at affordable prices (Khera, 2011). While food safety remains regulated through other laws, the Food Security Act places emphasis on availability, affordability, and nutrition, especially for economically vulnerable populations.

With regard to the definition of food, Sri Lankan law adopts a broad definition under section 33 of the Food Act, which defines food as any article manufactured, sold, or represented for use as food or drink for human beings. The definition does not expressly exclude items such as drugs, medicinal products, or agricultural products before harvesting. Indian food law provides greater clarity in defining food, and Indian courts have interpreted this definition in several cases. In State of Maharashtra v. Mohd. Yakub and similar decisions discussed in Indian jurisprudence, courts examined whether certain chewing products and processed agricultural items fell within the definition of food, relying on statutory inclusions and exclusions. These judicial interpretations support consistent application of food law in India, whereas Sri Lankan courts have had limited reported case law interpreting the scope of the definition of food.

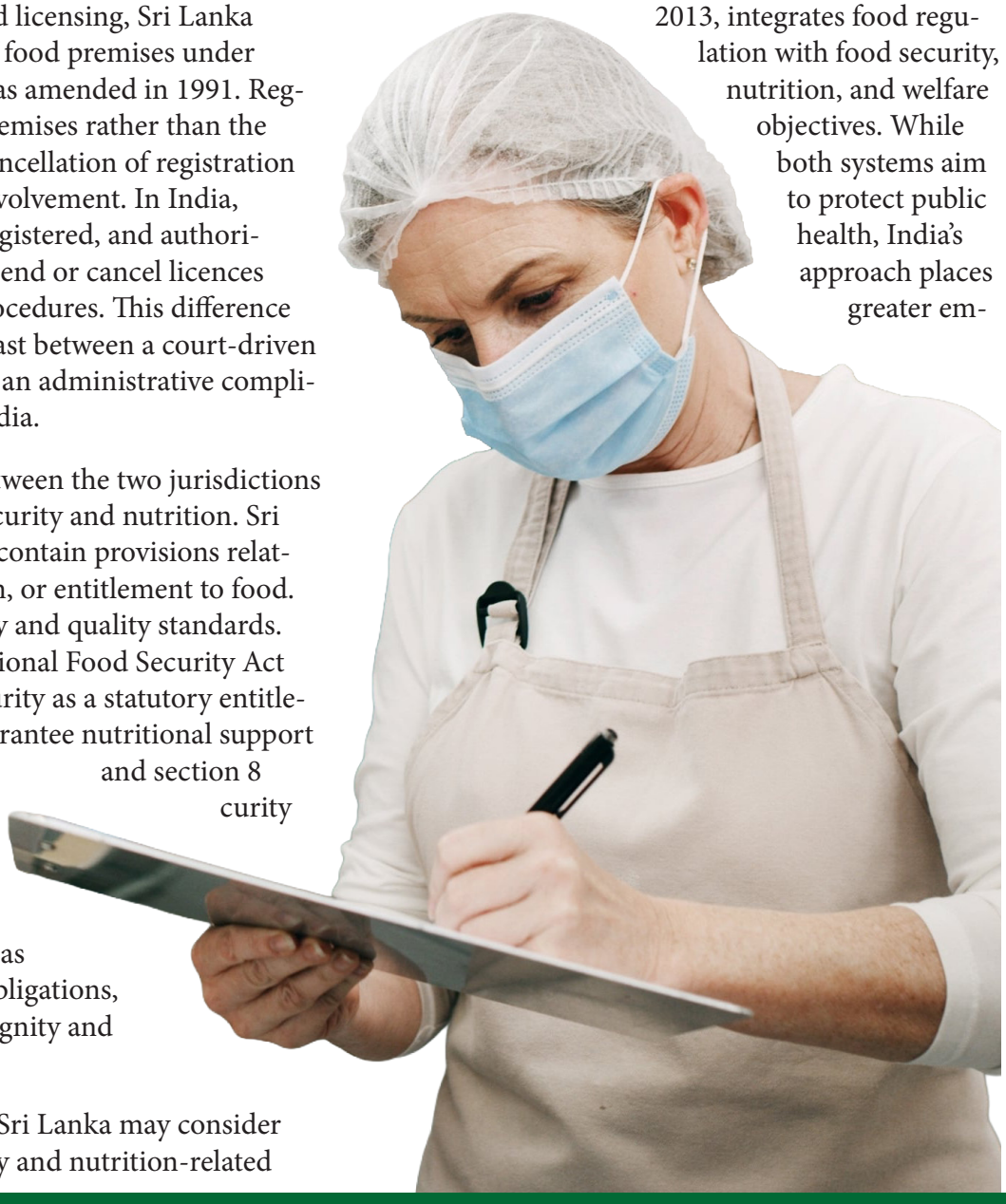
Institutionally, Sri Lanka's food law is administered mainly through the health sector. The Chief Food Authority, who is the Director-General of Health Services, oversees the administration of the Act. Local authorities such as Municipal Councils, Urban Councils, and Pradeshiya Sabhas function as Food Authorities, while Designated Officers such as Medical Officers of Health and Public Health Inspectors carry out inspections and prosecutions (Karunasena, 2010). India's system under the Food Security framework places responsibility on both central and state governments. The Act establishes clear obligations on governments and local authorities, supported by designated officers and food commissions that monitor implementation and address grievances (Patel and Nagar, 2014). This demonstrates that India has a more structured administrative framework, whereas Sri Lanka follows a health based enforcement model (Khera, 2011).

In terms of registration and licensing, Sri Lanka requires the registration of food premises under section 7 of the Food Act, as amended in 1991. Registration is linked to the premises rather than the food business itself, and cancellation of registration generally requires court involvement. In India, food businesses must be registered, and authorities have the power to suspend or cancel licences through administrative procedures. This difference reflects a procedural contrast between a court-driven approach in Sri Lanka and an administrative compliance-based approach in India.

A significant difference between the two jurisdictions is the treatment of food security and nutrition. Sri Lankan food law does not contain provisions relating to hunger, malnutrition, or entitlement to food. Its scope is limited to safety and quality standards. In contrast, the Indian National Food Security Act expressly reflects food security as a statutory entitlement. Sections 4 and 5 guarantee nutritional support for vulnerable groups, and section 8 provides for a food security allowance where entitled food is not supplied. Indian courts have identified these provisions as part of the state's welfare obligations, linking food access with dignity and social welfare principles.

Based on the comparison, Sri Lanka may consider incorporating food security and nutrition-related provisions into its legal framework, similar to sections 3 to 8 of the Indian National Food Security Act. Clearer statutory definitions of food and related terms could improve consistency in interpretation and enforcement. Administrative mechanisms such as grievance redressal officers and social audits may also be introduced to complement existing enforcement powers. Identifying vulnerable groups within food-related legislation could further strengthen food governance, drawing comparative lessons from the Indian jurisdiction.

The comparison between Sri Lankan and Indian food law demonstrates two distinct approaches to food regulation. Sri Lanka's legal framework focuses on food safety, quality control, and enforcement through inspections and prosecutions under the Food Act No. 26 of 1980 and its amendments (Hettiarachchi, 2021). India's framework, particularly under the National Food Security Act, 2013, integrates food regulation with food security, nutrition, and welfare objectives. While both systems aim to protect public health, India's approach places greater em-



phasis on access to food and institutional accountability, whereas Sri Lanka's approach remains on regulatory control and safety standards (Jayakumar, 2008).

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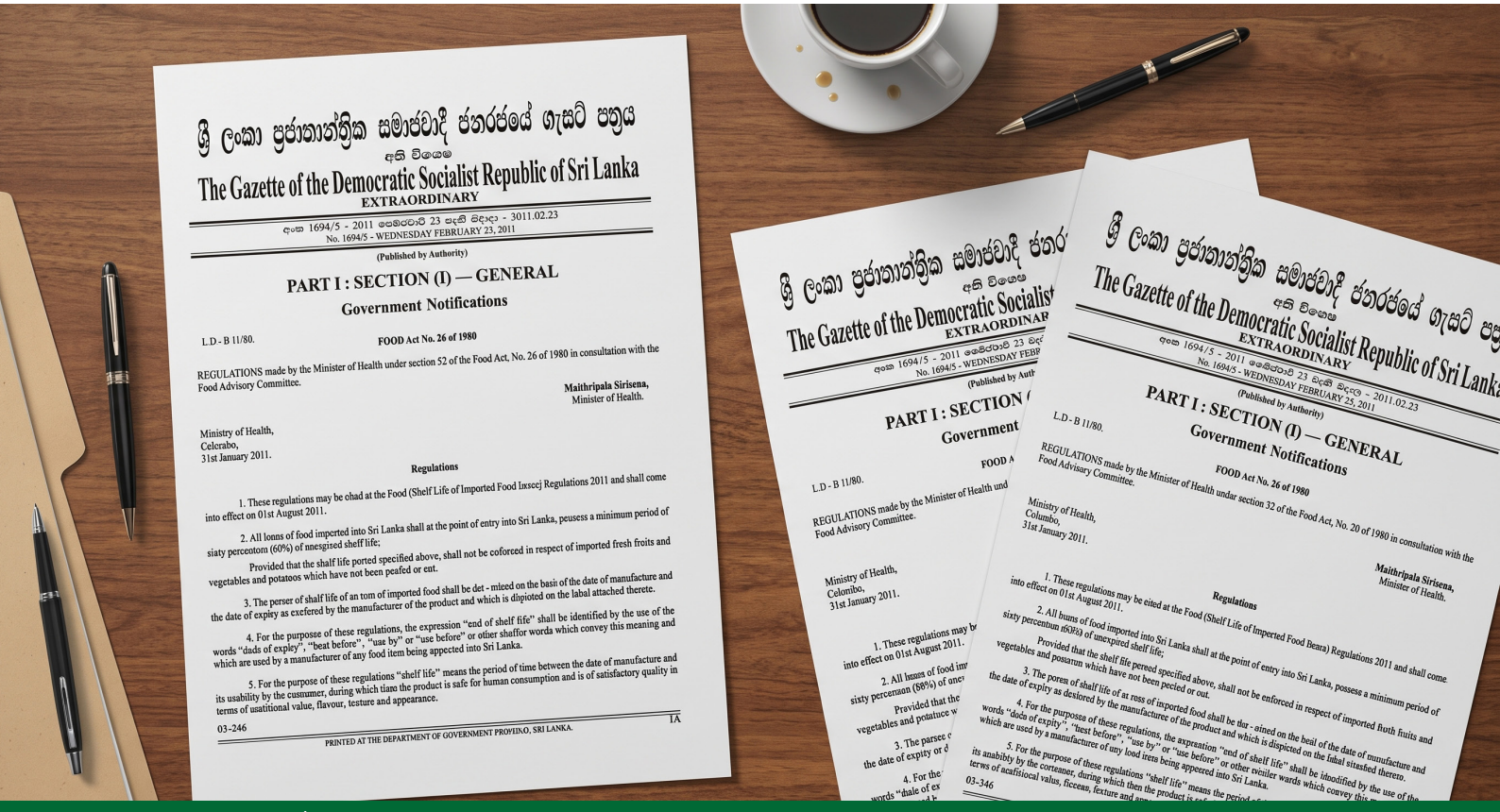

FROM FOREST SEED TO CHOCOLATE BAR

UNLOCKING MAHUA OIL'S COMMERCIAL POTENTIAL

The increasing demand for cocoa butter, driven by supply constraints and price fluctuations, has intensified efforts to develop novel, sustainable lipid alternatives for confectionery applications. *Madhuca longifolia* (mahua), a forest-derived oilseed commonly found in South Asian countries such as Sri Lanka, remains largely underexploited in modern food systems, despite its traditional use in household culinary practices. Recent research, however, has suggested its solid functional use as a fat ingredient with strong market potential.

Extraction efficiency of mahua oil is a critical first step toward its industrial utilization. In our first study, we demonstrated that ultrasound-assisted solvent extraction (UAE) offers a mild and efficient (energy usage 10 times lower than Soxhlet) extraction recovery of oil from mahua seeds compared to the conventional extraction techniques (Thilakarathna et al., 2023). Under mild operating conditions (35 min, 40% amplitude level, solvent-to-sample ratio of 25 mL/g, and 35 °C), UAE achieved an oil yield of approximately 57% with recovery exceeding 99%, comparable to Soxhlet extraction and substantially higher than that obtained by mechanical pressing (42.72%). Notably, UAE retained the native fatty acid composition of mahua oil, dominated by oleic (46%), palmitic (22%), and stearic acids (21%), along with a triacylglycerol (TAG) profile rich in Palmitoyl-Oleoyl-Olein (POO:22.9%), Palmitoyl-Oleoyl-Stearin (POS:17.4%), Stearoyl-Oleoyl-Olein (SOO:15.7%), and Palmitoyl-Oleoyl-Palmitin (POP:13.2%). Further, UAE-derived oil exhibited significantly greater antioxidant capacity (IC50 of DPPH 106.60 mg/mL and ABTS 39.80 mg/mL) than conventional methods (Soxhlet: IC50 of DPPH 810.40 mg/mL and ABTS 757.43 mg/mL)

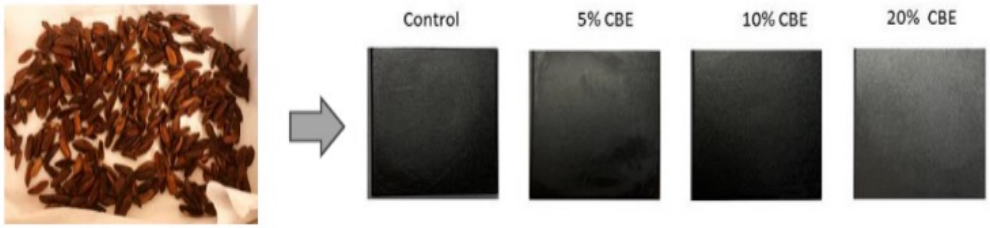

Figure 01: Mahua seeds Dark chocolates made from Mahua CBE

and Mechanical pressing (IC50 of DPPH 622.38 mg/mL and ABTS 392.87 mg/mL) (Thilakarathna et al., 2023).

Based on the potential TAG profile of mahua oil, our second study focused on modifying mahua oil into a cocoa butter equivalent (CBE) following a two-stage acetone fractionation process (Thilakarathna et al., 2024). This process selectively enriched symmetrical TAGs, yielding a final CBE collectively containing 76% of POP, POS, and SOS, meeting regulatory benchmarks for CBE. The resulting mahua CBE exhibited melting behavior closely aligned with cocoa butter (melting range near 18 °C), including a compatible crystalline structure (β_2 form) and a similar solid fat content. When mahua CBE was blended with Malaysian cocoa butter (10-90%), the mahua CBE showed up to 40% full compatibility, highlighting its practical suitability for confectionery applications (Thilakarathna et al., 2024). Interestingly, Mahua CBE closely resembles cocoa butter in its symmetrical TAGs, including proportions and the order of POS>SOS>POP. This approach reduces reliance on commonly used tropical fats such as shea, sal, and illipe, thereby positioning mahua oil as a locally adaptable and commercially scalable lipid alternative (Bahari & Akoh, 2018; Honfo et al., 2014).

"The increasing demand for cocoa butter, driven by supply constraints and price fluctuations, has intensified efforts to develop novel, sustainable lipid alternatives for confectionery applications. Madhuca longifolia (mahua), a forest-derived oilseed commonly found in South Asian countries such as Sri Lanka, remains largely underexploited in modern food systems, despite its traditional use in household culinary practices. Recent research, however, has suggested its solid functional use as a fat ingredient with strong market potential"

Analyzing the functional performance of the developed mahua CBE in dark compound chocolate systems is crucial for a novel fat alternative. In our third study, mahua POS-rich CBE was incorporated into dark compound chocolates at levels ranging from 5% to 40% (Thilakarathna et al., 2025). All formulations maintained acceptable particle size (D90: 10.9-18.1 µm) and rheological properties (Casson viscosity: 5.6-7.9 Pa; yield stress: 90-119 Pa), ensuring processability on standard equipment (Thilakarathna et al., 2025). Chocolates containing up to 20% CBE exhibited enhanced hardness during 12 weeks of storage, while higher substitution levels showed only modest softening. Notably, fat bloom, a major quality defect, was absent across all samples (low whitening index:18-21). Sensory testing revealed particularly strong consumer acceptance of chocolates containing 10% CBE, with hedonic scores comparable (6.9-7.4) control with 100% cocoa butter (Thilakarathna et al., 2025).

Conclusively, these findings establish a comprehensive pathway for the valorization of mahua seed oil, from efficient extraction and targeted lipid modification to its successful incorporation into chocolate products. While currently underutilized, mahua oil presents a promising, sustainable alternative to cocoa butter, offering both technological functionality and opportunities for regional value addition. Its transition from traditional household use to industrial applications may diversify fat sources, enhance supply resilience, and improve economic outcomes across the food sector.

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FROM SOIL NANOPARTICLES TO SUPER-FOOD PROPERTIES

The global market for functional foods is reported to be over 200 billion USD and is constantly on the rise due to the increasing demand for crops that are rich in phytonutrients. Some of the most commonly recognized examples of such foods are blueberries, which are rich in anthocyanins; kales, which are rich in glucosinolates; and quinoa, which is rich in protein. However, whereas most of the literature on the topic points towards the genetic makeup of the crop as a reason for its status as a "superfood," a new line of research is pointing towards a crucial factor that is not yet recognized: soil nanoparticles. Naturally occurring nanoparticles found in soil can be made up of a variety of compounds, including clay minerals, metal oxides, and mineral colloids.

Soil nanoparticles have sizes ranging from 1 to 100 nanometers. They have high surface area-to-volume ratios, which allows for effective interaction with plant root systems. Experimental studies have proven that nano-iron oxide plays a vital role in increasing iron uptake by wheat compared to other iron fertilizers (Al-Amri et al., 2020). The iron absorbed by plants from nano-iron oxide gets incorporated into protein fractions. Other nanoparticles regulate molecules and activate secondary metabolite biosynthesis pathways. Nano-zinc oxide plays a vital role in increasing phenylalanine ammonia lyase (Vatehovà et al., 2022), an essential enzyme for phenolic compound biosynthesis. Nano-zinc oxide increases phenolic compounds such as polyphenols, flavonoids, and anthocyanins. Nano-silica plays a vital role in increasing lycopene content in tomatoes by up to 45%. It also plays a vital role in increasing antioxidant capacity by over 60% (Alsaeedi et al., 2019). Unlike other methods of nutrient enhancement, which involve addition after harvesting, nanoparticles enhance nutrients during plant growth. This ensures stable bioactive compound integration into plant tissues.

The optimization of phytonutrients using nanoparticle-based interventions encompasses various classes of bioactive compounds. Nano-selenium application has been observed to increase the levels of glucosinolates in cruciferous vegetables such as Brassicas, thus increasing sulforaphane precursor levels with known cancer-preventive activity (Farouk and Al-Huqail, 2023). Nano-copper application has also been observed to increase carotenoid synthesis in vegetables such as carrots and sweet potatoes, thus increasing carotenoid levels with known provitamin A activity. Nano-titanium dioxide application in aromatic and

medicinal herbs has also been observed to increase essential oil yields through enhanced photosynthetic efficiency (Petry et al., 2014), thus increasing levels of bioactive monoterpenes.

What is Nano-Agriculture?

Nano-agriculture is an emerging field that applies nanotechnology principles to agricultural systems. It works in harmony with natural soil-plant interactions without genetic modification. Unlike genetic engineering, which still faces significant consumer skepticism, nano-agriculture does not alter the genetic composition of crops. Instead, it leverages the physicochemical properties of nanoparticles to modulate plant metabolism and nutrient uptake naturally.

How are Nanoparticles Used in Agriculture?

Nanoparticles are applied to agricultural systems primarily through soil amendment and foliar application. In soil amendment, nanoparticles are incorporated into the growing medium, where they interact with plant root systems to facilitate nutrient uptake. In foliar application, nanoparticle solutions are sprayed directly onto plant leaves, allowing absorption through the cuticle. These delivery methods ensure that nutrients are accumulated in plants during growth, resulting in greater bioavailability compared to nutrients applied as supplements post-harvest, as they become part of natural organic matrices with natural absorption enhancers.

Nutritional and Bioavailability Benefits

Nano-biofortification has significant potential for developing functional food products. Studies have shown that iron bioavailability in nano-biofortified beans is substantially higher compared to conventional bean varieties, thus illustrating the nutritional benefit of in-grown mineral nutrition. This approach offers a promising strategy for addressing micronutrient deficiencies while simultaneously enhancing the functional food quality of crops.

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FT 14

FROM WASTE TO WEALTH

THE EVOLUTION OF UPCYCLED FOOD SYSTEMS

The global food supply chain faces a major contradiction because one-third of all food produced for human consumption ends up lost or wasted, while food insecurity maintains a critical challenge. Traditionally, the food industry considered "food waste" as an essential by products that has been utilized from its processing activities. However, the upcycled food systems movement has developed new methods to create the valuable by products that prevent the accumulation of food waste from the food industries.

Upcycled food systems are defined as products containing various kinds of ingredients that have been wasted, which can be transformed through a value adding process into safe and high-quality food for human consumptions.

The Process of Valorization

Upcycling is more than just recycling; It is the process of Valorization, which transforms the discarded materials into valuable products. In the food industry, it often involves extracting bioactive compounds from agricultural waste products such as fruit pomace, spent grains, and vegetable peels. For instance, the brewing industry produces massive quantities of Brewers' Spent Grain (BSG) as waste material. Historically, BSG is mainly used for fed animals, but it is now processed into protein-rich flours and dietary fibers. Using upcycled food systems scenario does not just reduce waste; it enhances the nutritional profile of baked

goods through essential amino acid content and glycemic index reduction

Technological Frontier

Stepping into the circular economy concept, it relies on innovative extraction technologies in the food industry to establish and achieve the successful implementation of sustainable systems. Techniques such as Ultrasound-Assisted Extraction (UAE) and Supercritical Fluid Extraction (SFE) allow scientists to extract and isolate the sensitive antioxidants and polyphenol without using harsh chemical solvents and non- toxic extraction processes.

Ultrasound-assisted extraction (UAE): Uses sound waves to create cavitation bubbles, which break the plant cells and release the antioxidants without using high heat.

Supercritical Fluid Extraction (SFE): Helps to utilize CO_2 in a supercritical state, which isolates lipids and flavors without toxic residues. These "green" technologies ensure that safe ingredients, which remain clean label and consumer friendly.

Furthermore, the precision fermentation concept is mainly used for converting fruit sugars as feedstock from "ugly produce" such as fruit pulp and molasses into biopolymers and natural flavoring agents. As well as it produces high-value proteins or natural pigments as a byproduct. This system creates a closed-loop operation where the "waste" of one process becomes the "feedstock" for the next industry.

> **"The global food supply chain faces a major contradiction because one-third of all food produced for human consumption ends up lost or wasted, while food insecurity maintains a critical challenge. Traditionally, the food industry considered "food waste" as an essential by products that has been utilized from its processing activities. However, the upcycled food systems movement has developed new methods to create the valuable by products that prevent the accumulation of food waste from the food industries"**

Economic and Environment Impact

By utilizing the "Whole plant" manufactures can create new revenue methods from what was previously a disposal cost. Moving forward to economic achievement, the upcycling concept is very valuable. Environmentally, it helps to reduce the emission of methane associated with rotting in landfills and decreases the pressure and global warming on hydrosphere and geosphere needed to grow new cultivates.

The consumer Acceptance and Future

The success of Upcycle Food Systems heavily depends on consumer perception. Food Scientists need to establish the upcycled products by ensuring the safe for consumption and offer the high sensory characteristics of food products. Clear labelling and transparent manufacturing processes are essential for building consumer trust. With industry mature, the practice of Upcycling will become a common trend in manufacturing because it is addressed by both environmental needs and the economic potential from concealed resources.

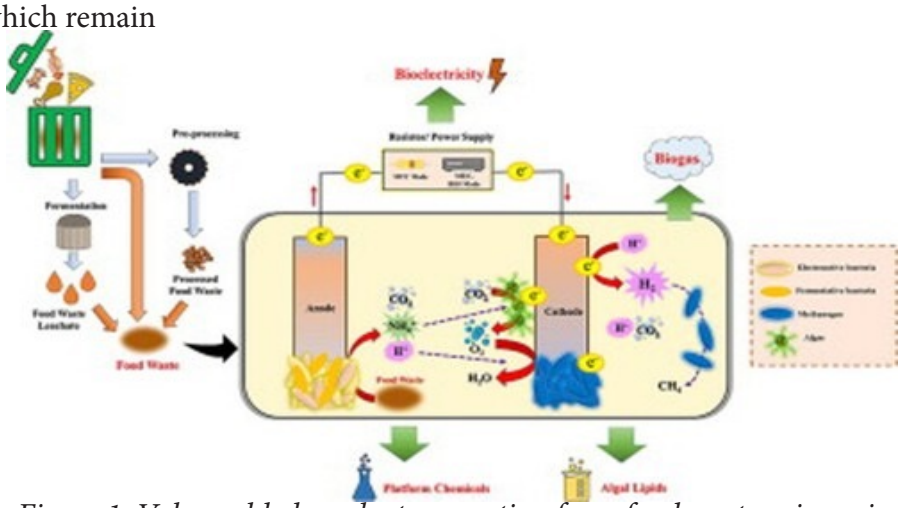

Figure 1: Value-added product generation from food waste using microbial process

As the increasing the global population, it approaches 10 billion, our current linear "take-make -dispose" model concept is no beneficial for environmentally sustainable and circular economy. In this current situation, Upcycling food Systems represents the sophisticated blend of traditional nutrition and modern emerging technology and biotechnology.

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FT 15

TINY GREENS, BIG IMPACT

THE FUTURE OF NUTRITION AND FOOD TECHNOLOGY THROUGH MICROGREENS

A Small Crop for a Big Food Challenge

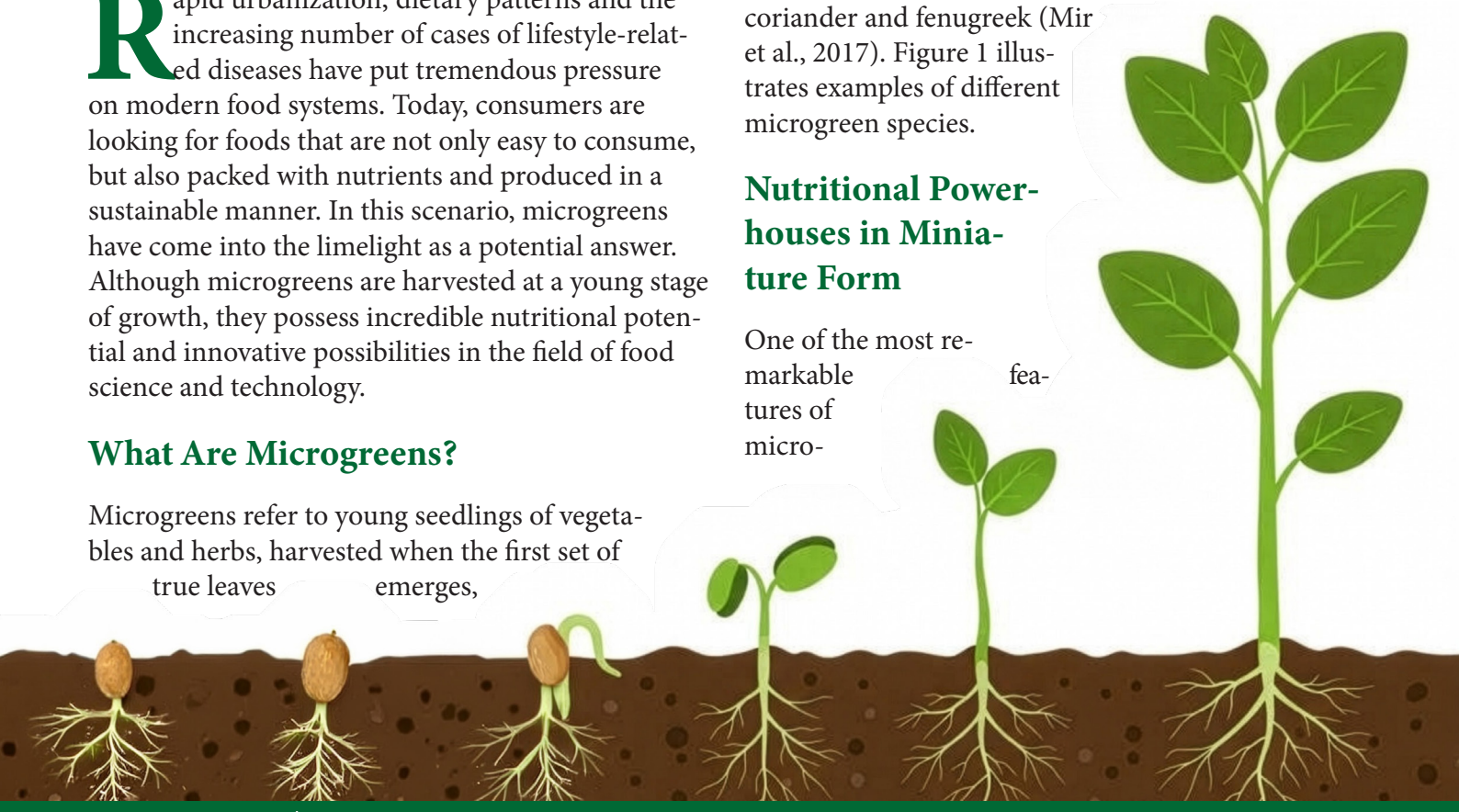
Rapid urbanization, dietary patterns and the increasing number of cases of lifestyle-related diseases have put tremendous pressure on modern food systems. Today, consumers are looking for foods that are not only easy to consume, but also packed with nutrients and produced in a sustainable manner. In this scenario, microgreens have come into the limelight as a potential answer. Although microgreens are harvested at a young stage of growth, they possess incredible nutritional potential and innovative possibilities in the field of food science and technology.

What Are Microgreens?

Microgreens refer to young seedlings of vegetables and herbs, harvested when the first set of true leaves emerges, usually between 7 to 21 days after germination (Kyriacou et al., 2020). Commonly grown varieties include broccoli, radish, mustard, beetroot, spinach, coriander and fenugreek (Mir et al., 2017). Figure 1 illustrates examples of different microgreen species.

Nutritional Powerhouses in Miniature Form

One of the most remarkable features of micro-



greens is their exceptional nutrient density, as research has proven that they contain higher levels of vitamins, minerals and natural components in foods that support health and help prevent diseases (Xiao et al., 2012). They are rich in vitamin C, vitamin E, carotenoids and phenolic compounds, making them powerful antioxidants. These characteristics classify them as functional foods, capable of reducing the risk of non-communicable diseases like cardiovascular diseases, diabetes and obesity, when consumed as part of a balanced diet (Singh et al., 2024). These nutritional qualities are associated with their harvest at a very early stage of plant development, as illustrated in figure 2.

Microgreens and Modern Food Technology

In terms of food science and technology, microgreens fit well within the technological advancements being made in controlled environment agriculture. Their short growth duration, less land requirement for production and less water requirement make microgreens an appropriate choice for hydroponic or vertical farming (Dubey et al., 2024). Technological innovations in nutrient solutions and LED lighting enable microgreens to be cultivated in a controlled manner. Studies have shown that microgreens could be optimized for phytochemical production by controlling light intensity and light quality. This would enable the nutritional enhancement of microgreens using technology (Kyriacou et al., 2020).

From Farm to Fork: Culinary and Commercial Applications

Apart from cultivation, microgreens are also gaining popularity in the food service and hospitality industries. The attractive look and strong taste of microgreens make them a great addition to burgers, sandwiches, wraps, pizza and other popular dishes like short eats, where they are used as nutritious toppings and fillings for the dishes served in the food industry (Xiao et al., 2012; Mir et al., 2017).

Microgreens can also be processed into powders and used as ingredients in soups, sauces and seasoning mixes, helping to create new food products while keeping their nutritional benefits (Seth et al., 2025). In addition, microgreens are becoming popular in urban food systems and professional kitchens because they support more sustainable food production and reduce the need for large-scale farming. From the food technology viewpoint, the use of microgreens in the preparation of ready-to-eat and value-added foods provide scope for the formulation of convenient foods with specific functions without affecting the taste and cooking behavior of the final product (Xiao et al., 2012).

Postharvest Challenges and Technological Solutions

Despite their advantages, microgreens are highly perishable due to their delicate tissues and high respiration rate. Rapid quality deterioration after harvest presents a major challenge (Singh et al., 2024). To address this, food technologists are exploring innova-

Figure 1: Photos of various microgreens (adapted from Zhang et al., 2021)

tive postharvest solutions, such as modified atmosphere packaging, edible coatings and biodegradable packaging materials. These technologies help to extend shelf life while maintaining nutritional quality and food safety, contributing to reducing food waste and improving sustainability (Mir et al., 2017).

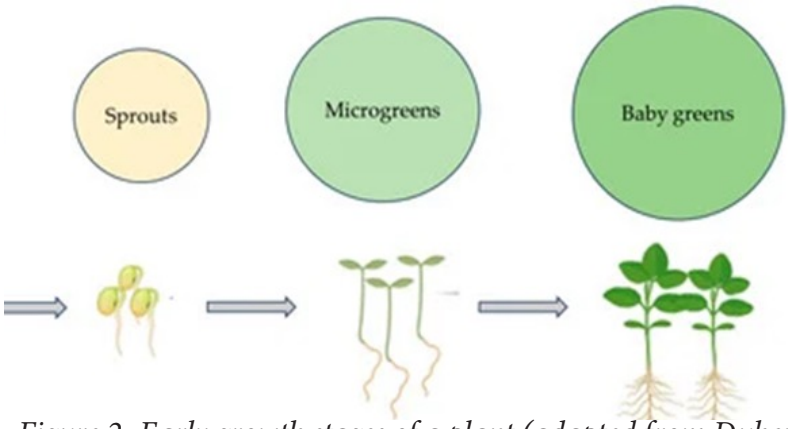


Figure 2: Early growth stages of a plant (adapted from Dubey et al., 2024)

Potential for Sri Lanka's Food System

In the Sri Lankan scenario, there is a potential for developing microgreens as an untapped market for urban farming and entrepreneurship, particularly in urban and semi-urban areas. In recent years, there has been a rise in home gardening, rooftop cultivation and sustainable food production. Therefore, microgreens can play a significant role in enhancing urban nutrition while creating income generating opportunities for small scale entrepreneurs and start-up agribusiness ventures. In Sri Lanka, crops such as Amaranthus, radish, fenugreek and leafy vegetables have the potential for being adapted for microgreens cultivation. However, there is a need for more awareness, research support and the incorporation of food technology to develop microgreens as a viable sector within urban farming systems and entrepreneurial agribusiness models.

Conclusion

The significance of microgreens is further highlighted by the fact that size is not an accurate measure of impact. Microgreens are nutrient-rich, easily integrated with cutting-edge technology, and are becoming an integral part of food services, thus making them a significant part of food systems in the future. As global and local challenges demand healthier and more sustainable food solutions, microgreens stand out as small greens with truly big potential.

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DIGITAL TWINS IN THE FOOD INDUSTRY

Introduction

A digital twin (DT) is a virtual model of a food processing operation, dynamically linked to its real-world counterpart via real- time sensor data and advanced analytics. This concept represents the convergence of key Industry 4.0 technologies, including the Internet of Things (IoT) sensor-driven data collection, big data analytics, artificial intelligence, and cloud computing. Unlike conventional simulation or monitoring systems which remain static after initial development, a digital twin continuously refines model predictions by updating its internal mathematical models and algorithms using live operational data. This ongoing refinement enables the system to generate increasingly accurate forecasts and deliver real time outputs that support process monitoring, management, and informed decision making.

How a Digital Twin (DT) Works?

A digital twin operates through continuous, bidirectional integration between a physical food production system and its corresponding virtual model. Sensors installed within the production environment collect operational data such as temperature, pressure, pH, flow rate, and product quality parameters. These data streams are transmitted to the digital model via Internet of Things (IoT) platforms and cloud-based infrastructure. The digital model then applies mechanistic models, such as Computational Fluid Dynamics for heat-transfer analysis, alongside data-driven machine learning algorithms to simulate process behavior, predict equipment performance, estimate product quality and identify process deviations .Critically, the virtual model not only receives data from the physical system but also feeds back optimization strategies and control signals, enabling real-time operational adjustments.This characteristic distinguishes DT from one-directional monitoring systems. The reliability of the digital twin depends on sensor data quality, a well-integrated multilayered modelling framework, computational performance and effective real-time synchronization with the physical process.

Three-tier Architecture

A food system digital twin is typically structured around three interconnected layers, the physical layer, where sensors (thermal, spectral, and smart

packaging sensors) monitor critical parameters such as temperature, pH, and water activity during processing or logistics; the cyber layer, where real-time data streams into cloud or edge computing infrastructures that integrate mechanistic models with machine learning (ML) algorithms for analysis and simulation; and the feedback layer, where the virtual model performs continuous, high-performance simulations to predict internal food states and automatically adjusts physical actuators or alerts operators to optimize process conditions. The overall architecture and data flow of a digital twin system for smart food processing are illustrated in Figure 1

Applications

1. Process Optimization and Fault Detection

In food processing and manufacturing, digital twins embed computational models directly into live production environments. This integration enables real-time fault detection, reduction of unplanned downtime, ongoing assessment of process performance, and dynamic optimization of processing conditions. Consequently, food process models transition from static design tools into active components of industrial operation, enabling adaptive parameter adjustment, improved production planning, and enhanced process consistency.

2. Food Quality, Safety and Sensory Reformulation

Digital twins can forecast quality and safety attributes that in-line sensors are unable to measure directly. By merging mechanistic models with real-time data, DT functions as virtual sensors to estimate variables such as temperature distribution, quality degradation, pathogen growth, spoilage progression, toxicity, and shelf-life. This predictive capability is particularly valuable in complex food systems with biological variability. Shelf-life prediction is achieved by modelling the microbiological and chemo-physical properties of foods and simulating the influences of food packaging interactions on perishability. In the dairy industry, for instance, DT-based models have been applied to predict microbial growth kinetics under dynamic storage conditions, while in the meat sector they have been used to simulate spoilage rates as a function of temperature and modified atmosphere packaging. In the realm of food reformulation

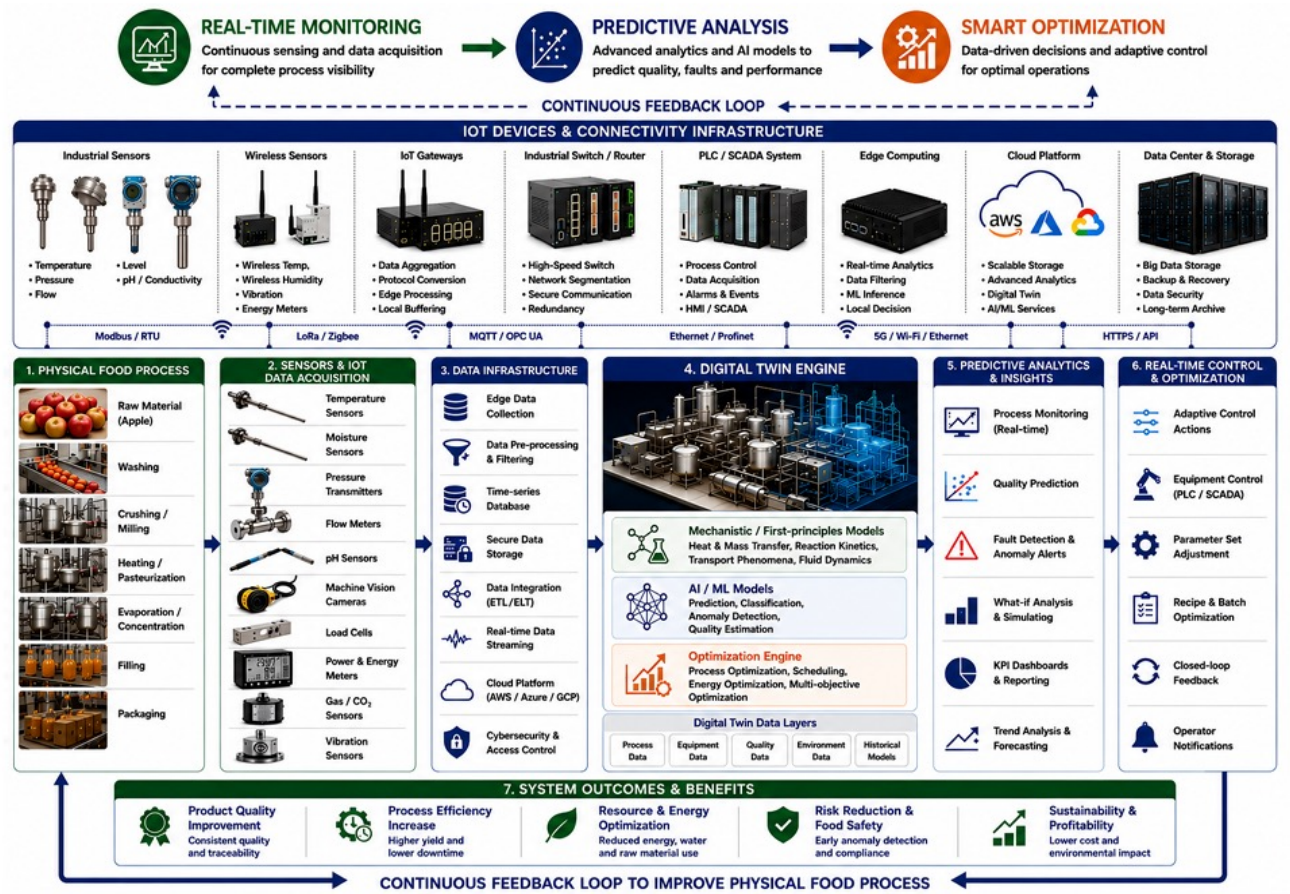

Figure 1: Architecture of an IoT-Enabled Digital Twin Framework for Smart Food Processing Systems. Source: Developed by the author based on published literature on (Verboven et al., 2020).

and product development, digital twins can predict how compositional changes, such as reducing fat, salt, or sugar or substituting animal proteins with plant-based alternatives, affect aroma perception. By combining physical models of aroma release with machine learning, DTs can predict aroma partition coefficients across foods with varying compositions. These models are applicable across diverse food categories, from dairy to plant-based products, supporting the creation of nutritionally enhanced foods while preserving sensory quality.

3. Supply Chain Resilience and Traceability

Digital twins strengthen the resilience of food supply chains by simulating disruption scenarios, such as cold chain failures, and enabling proactive risk management. By integrating sensor-derived data with machine learning, they facilitate real-time monitoring and adaptive adjustments across production and logistics networks. Smart packaging equipped with embedded sensors further enhances data collection and visibility throughout the supply chain. Digital twins also improve food traceability. When integrated with blockchain technology, digital twins provide immutable, transparent records of ingredient origins, processing conditions, and distribution history. Mechanistically, the digital twin converts real-time sensor data, including ingredient provenance, processing parameters, and cold-chain temperatures, into secure cryptographic records stored on a distributed blockchain ledger. This decentralized framework creates an immutable and tamper-resistant history of the product throughout the supply chain, preventing unauthorized data modification. Consequently, the shared ledger enables transparent cross-organizational access among producers, processors, logistics providers, and regulators, establishing a reliable and verifiable source of information across the food value chain. This combination enables rapid product recall management, supports regulatory compliance, and builds consumer trust by offering verifiable provenance data at each stage of the food production lifecycle.

Advantages and Limitations

The adoption of digital twin technology in food systems offers a broad range of operational and strategic benefits. These include reduced product variability and improved consistency, enhanced food quality, safety, and extended shelf life, as well as minimized food losses and waste. Furthermore, digital twins contribute to decreased energy and resource consumption, optimised maintenance scheduling through predictive fault detection, improved production planning and logistics optimisation, and greater supply chain transparency and traceability. However, several significant challenges constrain wider adoption. The technology demands high model complexity and multidisciplinary expertise spanning food science, engineering, and data science, while integrated, commercially viable digital twin platforms remain limited within the food sector. Substantial computational demands require high-performance cloud or edge computing infrastructure, and in-line sensor coverage for all relevant process variables remains insufficient in many settings. Difficulties in data sharing across supply chain partners, driven by proprietary concerns and interoperability gaps, further hinder progress. Additionally, cost implications vary considerably depending on the scale and complexity of the application, and challenges surrounding model validation, scalability, and demonstrating clear economic feasibility continue to present barriers to widespread implementation.

Conclusion

Digital twins represent a transformative advancement in modern food process modeling, moving beyond static simulation to provide continuous, real-time decision support throughout the food production lifecycle. By dynamically linking physical systems with their virtual counterparts, digital twins enable proactive quality management, predictive safety monitoring, and adaptive supply chain optimization, capabilities that are increasingly critical in a rapidly evolving global food system. Realizing the full potential of digital twin technology in food systems requires sustained multidisciplinary collaboration across food science, engineering, data science, and regulatory domains. It also demands investment in robust sensor networks, high-performance computing infrastructure, and industry-wide data sharing frameworks. As these enablers mature, digital twins are poised to become a foundational technology for next-generation food production, delivering measurable improvements in efficiency, safety, sustainability, and transparency.

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FT 17

FROM ROOT TO RISK

THE STORY OF LINAMARIN IN CASSAVA

Cassava (*Manihot esculenta*) is a major staple food crop for over 500 million people worldwide, especially in tropical and subtropical Africa, Asia, and Latin America, due to the high carbohydrate content and the ability to cultivate under adverse climatic conditions. As mentioned by the United Nations Food and Agriculture Organization (FAO), cassava is the fourth-ranked food crop after rice, maize, and wheat in developing countries. Cassava root contains 25-30 % starch and other valuable nutrients like potassium, iron, calcium, vitamin A, folic acid, sodium, vitamin C, vitamin B-6, and protein. The concern is that cassava contains linamarin, a cyanogenic glycoside (CNG) that can release toxic hydrogen cyanide (HCN), which may cause severe health issues. So, adequate processing and careful control of linamarin during processing is essential to ensure food safety and public health (FAO/WHO, 2012).

Linamarin is the most prominent glycoside in cassava, which is primarily stored in vacuoles in cassava cells. Cassava roots and leaves account for the higher linamarin content in cassava. Linamarase, naturally present in cassava, is the enzyme responsible for its hydrolysis, which ultimately produces HCN. In normal physiological conditions, linamarase enzyme is spatially sepa-

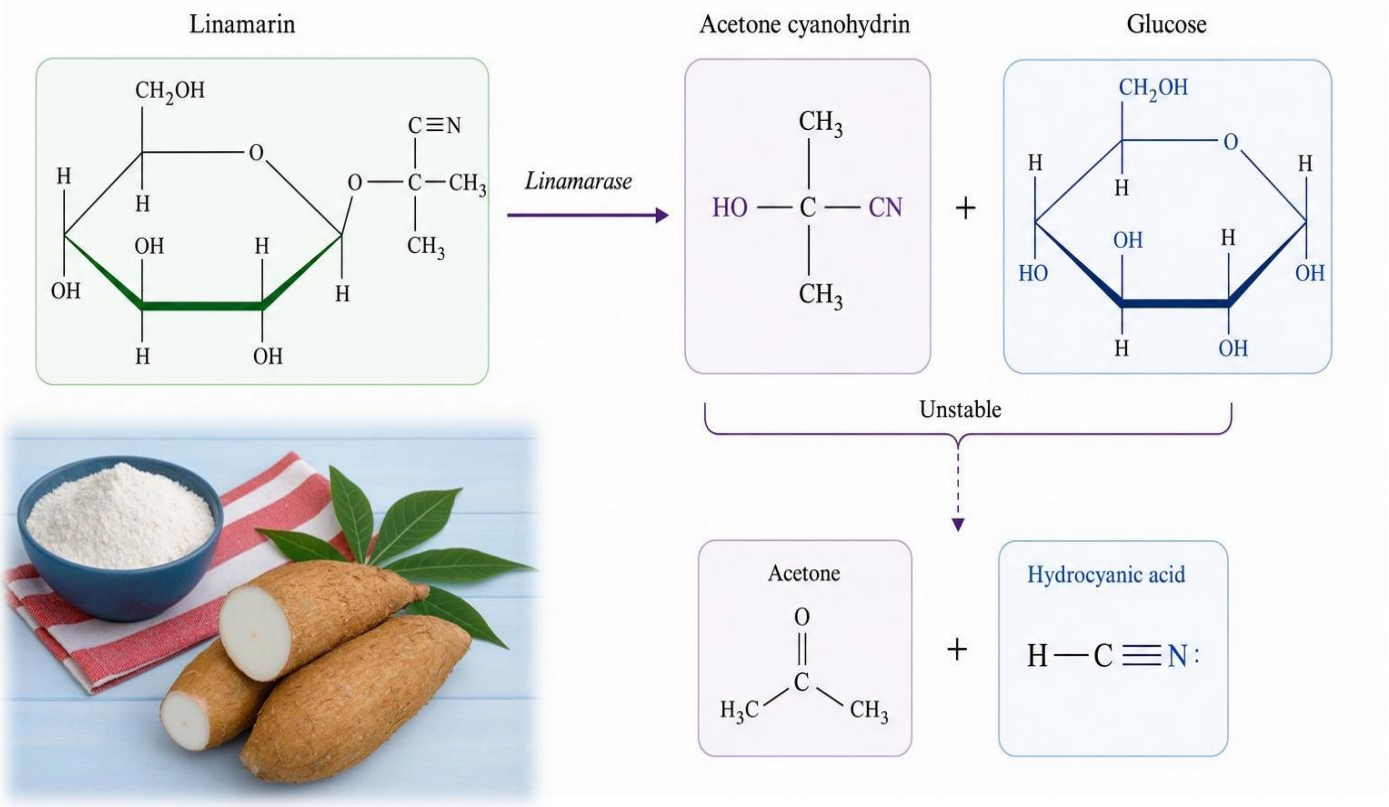

Figure 01: Hydrolysis of linamarin by linamarase ultimately producing HCN.

rated from linamarin by being located in the cell wall and cytoplasm. When tissues are damaged during harvesting, transporting or processing, linamarin comes into contact with linamarase. The linamarase catalyzes the hydrolysis of linamarin to acetone cyanohydrin and glucose. Acetone cyanohydrin is unstable and rapidly breaks down to produce hydrocyanic acid, especially at higher temperatures and neutral to alkaline pH. This formed cyanohydrin decomposition is less in low pH conditions (Figure 01).

Cyanide can inhibit the cytochrome oxidase enzyme and prevent the cells from using oxygen to produce energy, interfering with cellular respiration. Consuming lower cyanide levels is not lethal, but long-term intake of even lower levels may cause severe health issues such as neurological disorders, glucose intolerance, goitre due to impaired iodine metabolism, and a paralytic disease known as konzo. Symptoms of acute toxicity occur within 4 hours, including headache, dizziness, nausea, vomiting, abdominal pain, rapid breathing, and weakness, while severe poisoning may result in respiratory failure and can be fatal (FAO/WHO, 2012). Some are suggested intake of over 20 mg per 100 g of cassava is toxic, while some set the toxic level at an intake of 50 to 60 mg daily for a European adult. 15 and 400 mg HCN/kg fresh weight is the normal range of cyanogen content of cassava tubers.The toxicity occurs only due to the hydrolysis of linamarin. If the intestine absorbs linamarin in its intact

glycoside form, it is excreted unchanged via the urine without causing any toxic effect. However, during digestion, this ingested linamarin can be broken down by gut enzymes to produce cyanides.

For the communities that consume cassava as the major staple, cassava toxicity can be prevented by consuming cassava varieties that have lower cyanogenic content or by using proper processing methods to reduce cyanogenic content up to a safer level, even in high cyanogenic content varieties. By applying traditional and industrial processing techniques like peeling and slicing or grating of fresh cassava roots, followed by soaking, boiling, drying, and fermenting, the cyanide content can be reduced to a safer level. Peels of cassava roots contain a higher amount of cyanide, and by peeling off, the cyanide content can be lowered. Grating or crushing will facilitate the linamarase enzyme to produce HCN, and then the produced HCN is removed using subsequent steps, such as soaking and fermentation, which facilitates the leaching of cyanide into the water.

When cooking cassava tubers, cut into small pieces, up to 80% CNG content is leached into the water, but it is necessary to decant the water completely. Also, when boiling or cooking, the vessel should be opened

to facilitate the evaporation of HCN. Processing methods such as baking, steaming, and frying lead to a minor loss of CNG (20%) due to the destruction of linamarin and the high stability of linamarin at higher temperatures. Cassava chips, around 10 mm in thickness, have shown 80% of reduction of CNG content due to the enzymatic activity of linamarase and extended drying time rather than thermal effect. Grating before sun drying is most effective because the grating enhances the enzymatic activity by further increasing surface area, resulting 95%-99% reduction of linamarin content.

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MEDIA MYTHS VS REAL FOOD SCIENCE

In today's digital world Media plays a big and powerful role in shaping public perceptions of food and nutrition. Social media platforms, news outlets, and blogs often simplify or sensationalize scientific findings, giving rise to widespread food myths that conflict with established food science. Understanding the difference between media myths and evidence-based food science is essential for making informed dietary choices.

> **"Processing isn't the enemy — pasteurization, fortification, and freezing often make food safer, not less nutritious"**

Influencers and self-proclaimed "gurus" frequently spread misleading, attention grabbing stories about food and diets-amplified by algorithms that favor controversy over accuracy. While studies show that myths persist even among educated groups, such as future secondary school teachers. Academic training alone does not always protect them from believing common myths, and many still pick up nutrition ideas from media, the internet, family traditions, and social networks rather than from original scientific literature.

New food technologies like Genetically Modified (GM) foods are particularly vulnerable to simplified media frames and affect-based judgments rather than detailed understanding One of the most common media myths is that "Only natural is always safer, healthier, and more nutritiou than scientifically developed foods". Media discussions often favor only "natural" foods while raising doubts about Genetically Modified Organisms (GMOs). However, food science evaluates the safety of products through rigorous testing andregulatory approval processes. Research shows that GM crops, such as insect-resistant Bt crops, golden rice, and herbicide-tolerant varieties can enhance nutritional value, improve resistance to pests, and contribute to food security and sustainability.

Another widespread myth is that "processed foods are always unhealthy." In reality, processing can include beneficial steps like pasteurization (which significantly reduces microbial risks), fortification, and freezing often improving safety, shelf life, and even

nutritional value. Yet media and consumers frequently lump all processing together as "junk food".

Cultural beliefs also play a role; for example, in Sri Lanka, many people believe "brown or village eggs are more nutritious than white or commercial eggs." While in Europe and Brazil many citizens still hold strong misconceptions about functional foods and biotechnology. These misconceptions directly shape attitudes and purchasing decisions.

To combat myths effectively, research highlights several strategies: nutrition professionals should actively model evidence-based thinking and transparent practice. "Truth sandwich" texts significantly reduce agreement with popular nutrition myths, while warning people in advance about common misinformation tactics is more effective than correcting myths after exposure. When people are asked which sources they consider most reliable for diet and health decisions, they consistently report higher trust in nutrition scientists, qualified health professionals, universities, and scientific journals than in celebrities, influencers, or general social media content. In practice, however, people often use quick and popular online sources more frequently than expert sources, which helps explain why myths remain widespread despite this stated trust in scientific and professional authorities.

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FT 19

MICROBIAL PROTEINS

THE SUSTAINABLE PROTEIN FRONTIER BEYOND ANIMAL AND PLANT SOURCES

Traditional food systems are feeling the pressure like never before as the world's population grows towards an expected 9.7 billion in 2050. Conventional livestock production, though an important part of the human diet, is also one of the main contributors to environmental degradation, including greenhouse gases, land degradation, and water pollution. Although plant- based proteins are often promoted as sustainable alternatives, they also face limitations such as lower digestibility, imbalanced amino acid composition, anti-nutritional compounds, and high agricultural land dependency. But in response to these challenges, a quiet revolution is simmering in bioreactors across the globe, 'alternative proteins through fermentation'. This old technology is being used to produce a new generation of sustainable, healthy, and ethical animal and plant protein alternatives

Why microbial proteins are important beyond animal and plant proteins

This process harvests high protein yield within short production cycles with minimum requirement of land and water. Microbial protein production requires a fraction of the land and water needed for traditional ranching methods. For example, producing 1 kg of beef may result in emissions of between 27 and 60 kg of CO2 equivalent, while fermentation- based protein production may use a fraction of this carbon footprint (Cedeno, Olubiyo, & Ferreira, 2025). In addition, fermentation technologies may utilize waste such as molasses and cheese whey as feedstock for protein production, thus utilizing waste as a valuable resource for protein production. Microbial proteins are not affected from climate and seasonal variation. There is an improved amino acid composition and digestibility in microbial proteins and also naturally free from hormones and antibiotics.

Modern cellular agriculture mainly utilizes three complementary fermentation approaches for microbial protein production

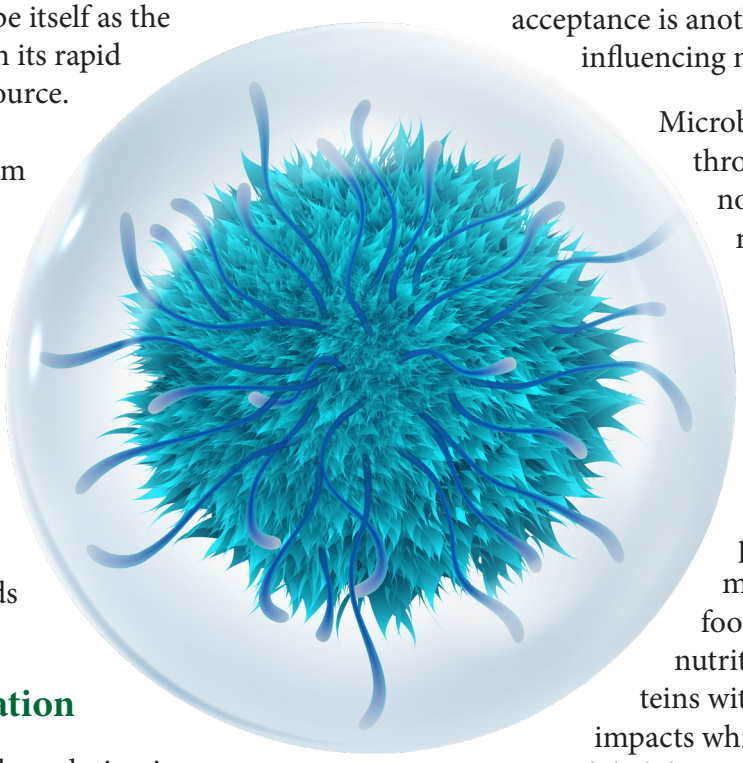
1.Traditional Fermentation

This involves the use of live microbes to alter the chemical properties of plant materials. In the alternative protein industry, traditional fermentation is an effective method of enhancing the nutritional value

of legumes and grains by reducing anti- nutrients such as phytic acids and tannins, which can otherwise inhibit the absorption of nutrients. For instance, the fermentation of lupin seeds using yeast and lactic acid bacteria has been demonstrated to enhance the digestibility of protein to levels similar to those of soybean meal (Kårlund, et al., 2020).

2.Biomass Fermentation

It is centered on the microbe itself as the food source, particularly on its rapid proliferation as a protein source. The best-known product is mycoprotein, produced from fungal biomass such as Fusarium venenatum. These proteins are not only easily scalable but also possess a meat- like texture and a strong nutritional profile that low in saturated fat and higher fiber. Mycoprotein has a protein content of 50- 55% and is rich in essential amino acids such as lysine.



3.Precision Fermentation

This is the fourth industrial revolution in food. Precision fermentation technology employs 'cell factories', genetically engineered microorganisms to produce targeted, high value functional ingredients that are bio identical to animal proteins. From recombinant whey and casein for non- dairy milk products to heme proteins such as myoglobin, which gives plant- based burgers their taste and appearance, precision fermentation enables the production of products that have the same exact sensory and nutritional attributes as their animal- derived counterparts without the need for animal agriculture.

Commercial landscape and future outlook

The commercial environment for fermented proteins is growing at a very fast pace. Companies like New Culture are already producing bovine ß- casein using precision fermentation to make mozzarella cheese that melts and has the same taste as traditional cheese. In the mycoprotein category, brands such as Quorn have already opened the door for new entrants such as Mycorena and Aqua Cultured Foods, who are working on seafood alternatives made from fungi.

There are still some issues to be addressed. The current need is to scale up these processes from the lab to commercial production. The regulatory environment is also struggling to keep up with the technology, especially in terms of safety and GMO (Genetically Modified Organisms) labeling. Consumer acceptance is another important factor influencing market expansion.

Microbial proteins produced through fermentation are no longer considered niche innovations. Instead, they represent a scientifically advanced and environmentally sustainable protein source capable of complementing both animal and plant proteins. By harnessing microbial systems, the food industry can produce nutritionally balanced proteins with lower environmental impacts while supporting the growing global demand for sustainable food production. As biotechnology continues to advance, microbial proteins are likely to become an essential component of future food systems.

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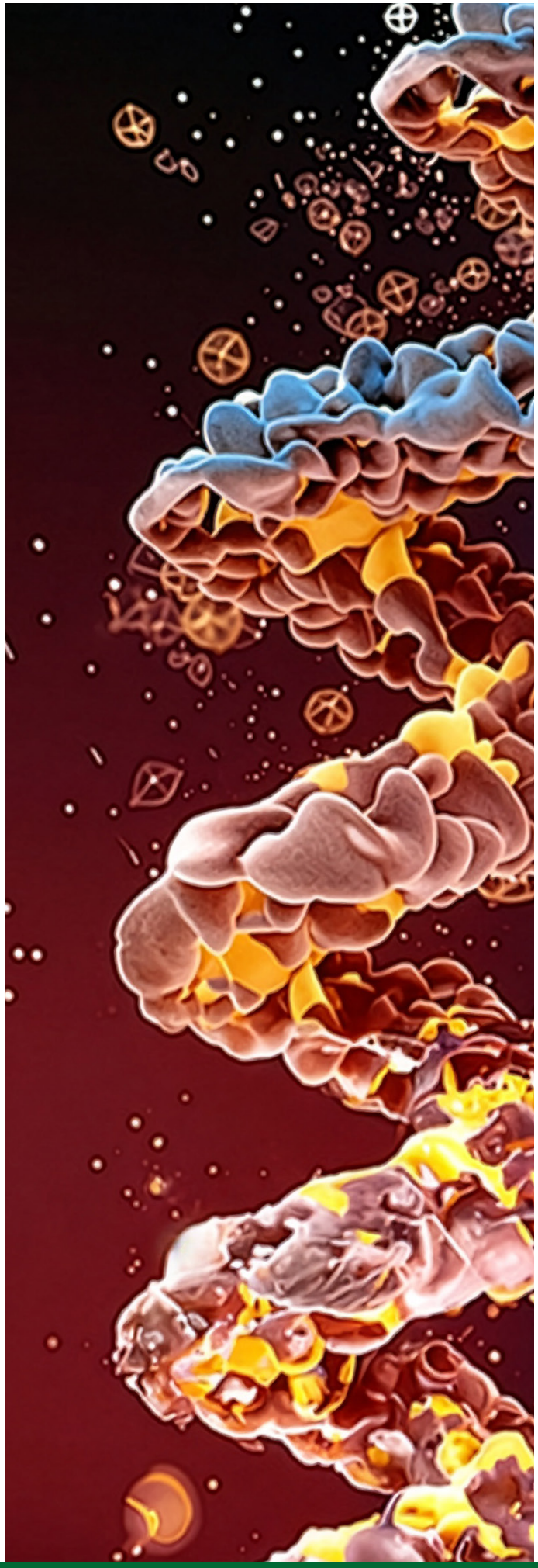

FT 20

ULTRA-PROCESSED FOOD

CONVENIENCE AT A COST

The global production and consumption of processed foods have increased rapidly over recent decades. In 2020, the global market for processed foods was estimated at about USD 1.93 billion and is expected to rise to nearly USD 3.41 billion by 2030. This growth reflects changing lifestyles driven by urbanization, increased disposable income, and the expansion of modern food retail systems. Processed foods also contribute positively to food systems by extending shelf life, reducing post-harvest losses, and improving food availability.

However, growing evidence links excessive consumption of ultra-processed foods (UPFs) with adverse health outcomes. Diets high in fats, sugars, and salt-combined with sedentary lifestyles are strongly associated with type 2 diabetes, obesity, cardiovascular diseases, hypertension, and other non-communicable diseases. Aggressive marketing strategies, particularly those targeting children through emotional appeals and misleading health claims, further promote overconsumption and undermine traditional dietary patterns, disproportionately affecting vulnerable populations.

NOVA Food Classification System: Four Food Groups

The NOVA system classifies foods based on the degree and intent of processing, instead of relying solely on nutrient-based claims. It distinguishes between beneficial and harmful forms of processing and includes four groups.

Essential Features of Ultra-Processed Foods (UPFs)

UPFs are products of modern industrial food systems designed mainly for convenience, high palatability, and commercial profitability. Their characteristics reflect profound changes in food composition, structure, and consumption behavior rather than simple processing.

Table 1. Key aspects of NOVA food classification Table

NOVA Group	Definition	Examples	Purpose of Processing	Nutritional Impact
Group 1: Unprocessed / Minimally Processed Foods	Natural foods that undergo minimal processing to remove inedible parts or ensure safety, without major alterations.	Fresh fruits, vegetables, meats, grains, nuts	Preserve, store, make palatable	Varied nutrients, often fiber-rich, essential for a balanced diet
Group 2: Processed Culinary Ingredients	Substances derived from natural foods through methods like pressing or refining, used in meal preparation.	Oils, butter, sugar, salt, starch	Used to prepare, season, and cook meals	High in energy, often nutrient-poor, used to enhance taste and texture
Group 3: Processed Foods	Foods with added ingredients from Group 2 for preservation, flavor, or sensory qualities. Usually recognizable as modified Group 1 foods.	Canned/ bottled fruits & vegetables, smoked fish, fresh cheeses, breads etc.	Extend shelf life, enhance taste	Moderate energy density, can be nutritionally balanced in small amounts
Group 4: Ultra - Processed Foods	Formulations with little or no whole foods, made through extensive industrial processing, often with additives for flavor, texture, and shelf life.	Soft drinks, packaged snacks, instant noodles, confectionery, etc.	Increase shelf life, improve palatability, marketability	Low nutritional quality, often high in unhealthy fats, sugars, and salt; associated with poor diet quality

1. Complex Ingredient Composition

UPFs typically contain long ingredient lists, including refined sugars, modified starches, hydrogenated or interesterified oils, preservatives, colorants, emulsifiers, and flavor enhancers. These additives improve texture, shelf life, flavor, and appearance but often replace naturally occurring nutrients and bioactive compounds. Consequently, UPFs are usually energy-dense and micronutrient-poor.

2. Extensive Industrial Processing

UPFs undergo multiple processing steps such as refining, extrusion, fractionation, and chemical modification, which disrupt the natural food matrix. This structural breakdown reduces nutrient integrity, alters glycemic responses, lowers satiety, and facilitates rapid calorie absorption, increasing the likelihood of overeating.

3. Low Content of Whole Foods

Whole or minimally processed foods are either absent or present in small amounts, often as concentrates or isolates. This reduces dietary fiber and protective phytochemicals, contributing to nutritionally imbalanced diets with limited diversity and health-promoting components.

4. Sensory Optimization and High Palatability

UPFs are engineered to maximize palatability through carefully balanced combinations of sugar, salt, fat, and appealing textures. This sensory design can override appetite regulation, promoting rapid consumption, frequent snacking, and excessive intake, particularly among children and adolescents.

5. Convenience and Long Shelf Life

UPFs are widely available, ready-to-eat, and shelf-stable due to preservatives and protective packaging. While convenient for modern lifestyles, frequent reliance on these foods displaces home-prepared meals and traditional dietary practices, reinforcing unhealthy eating patterns.

> **“Anne Wigmore’s quote reminds us that everyday food choices matter. In an era where ultra-processed foods dominate shelves and meals, convenience often comes at the cost of nutritional quality and long-term health”**

Health Implications: What the Evidence Shows

fects on gut microbiota, and low-grade inflammation caused by certain additives and highly refined ingredients.

Moving Toward Healthier Food Systems

Complete avoidance of ultra-processed foods is unrealistic; however, reducing reliance on them is both feasible and beneficial. Practical strategies include prioritizing whole or minimally processed foods, reading ingredient labels carefully, and supporting

Table 2. Meta-analyses reporting associations between UPF consumption and health implications

Health Implication	Risk increment	Reference
Type 2 diabetes	40% increased risk	Chen et al (2023)
Cardiovascular events	35% increased risk	Yuan et al (2023)
Hypertension	23% increased risk	Wang et al (2022)
All-cause mortality	29% increased risk	Taneri et al (2022)
Overweight	36% increased risk	Moradi et al (2021)
Obesity	55% increased risk	
Metabolic syndrome	81% increased risk	Lane et al (2021)
Cardiovascular mortality	50% increased risk	Suksatan et al (2021)
Cardiac mortality	66% increased risk	

Extensive research findings and meta-analyses link high consumption of UPFs with adverse health outcomes. High consumption of UPFs is consistently associated with poor dietary quality, characterized by increased intake of free sugars, saturated fat, and sodium, and reduced intake of fiber, vitamins, and essential minerals. Epidemiological studies link UPF-rich diets to higher risks of type 2 diabetes, obesity, hypertension, metabolic syndrome, cardiovascular diseases, depression, and increased all-cause mortality. Proposed mechanisms include disruption of appetite control, excessive energy intake, adverse effood innovations that emphasize nutrition, sustainability, and transparency. At the policy level, clearer labeling, restrictions on marketing to children, and supportive food environments are essential.

Key Takeaway

The NOVA classification highlights that food processing matters as much as food composition. While ultra-processed foods offer convenience, their widespread consumption poses clear health risks. Shifting dietary patterns toward minimally processed foods,

supported by informed consumer choices and effective policies, is critical for improving public health and ensuring sustainable food systems.

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DRIVING INNOVATIONS AND VALUE ADDITION IN MODERN DAIRY PRODUCTS

Enzyme technology has emerged as a powerful driver of innovation in the modern dairy industry, enabling the development of products with improved texture, flavor, nutritional value, and processing efficiency. As natural biocatalysts, enzymes offer highly specific and environmentally friendly solutions for transforming milk into a wide range of value-added dairy products, including cheese, yoghurt, lactose-free milk, dairy beverages, and functional foods.

Among the most widely applied enzymes in dairy processing are chymosin, proteases, lipases, lactase, and transglutaminase. These enzymes play crucial roles in milk coagulation, protein and fat modification, lactose hydrolysis, and structural enhancement of dairy matrices. Their targeted action allows manufacturers to tailor product characteristics while maintaining consistency and quality.

Enzyme Applications in Cheese Production

Cheese production remains the most established application of Enzyme Technology in the dairy industry. Chymosin, the primary milk-clotting enzyme, initiates curd formation by cleaving κ-casein, resulting in controlled coagulation and desirable curd firmness. Microbial and recombinant chymosin are increasingly used to ensure consistent performance and to meet vegetarian preferences.

During cheese manufacture and ripening, naturally occurring milk enzymes and microbial enzymes produced by starter cultures contribute to the breakdown of proteins. Among these, proteases play a critical role by hydrolyzing casein into peptides and free amino acids, leading to texture softening and the development of complexed in the production of cheeses like Cheddar, Mozzarella, and Blue cheese to improve coagulation efficiency, accelerate ripening, and enhance flavor development, flavor compounds during maturation. Lipases further enhance cheese quality by hydrolyzing milk fat into free fatty acids, esters, lactones, and methyl ketones, producing distinctive buttery, fruity, and sharp flavor notes essential for aged and enzyme-modified cheeses. In industrial dairy processing, enzymes such as chymosin and lipase are widely applied in the production of cheeses like Cheddar, Mozzarella, and Blue cheese to improve coagulation efficiency, accelerate ripening,

and enhance flavor development.

Lactose Hydrolysis and Functional Dairy Products

Lactose hydrolysis is a major value-adding application of Enzyme Technology in the dairy industry.Lactase (β-galactosidase) breaks down lactose into glucose and galactose, improving digestibility for lactose-intolerant consumers while enhancing the natural sweetness of dairy products.

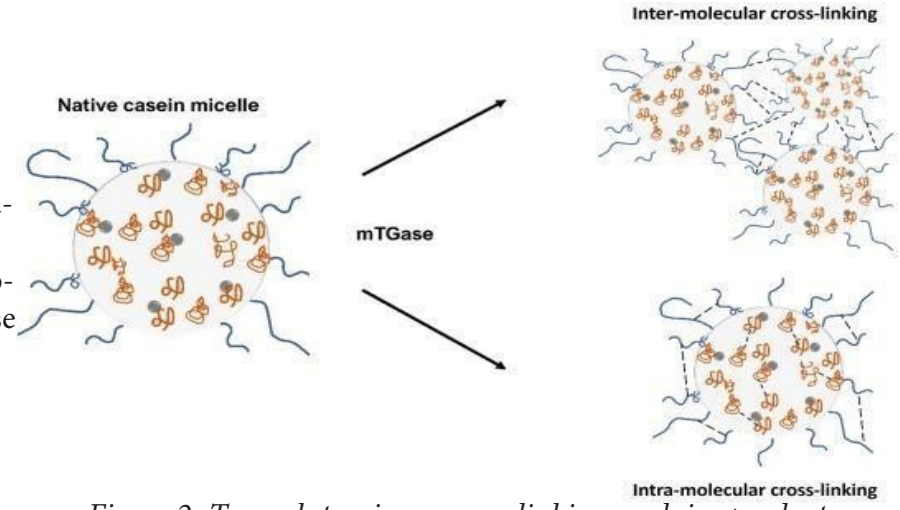

Figure 2: Transglutaminase cross-linking on dairy product (Velazquez-Dominguez et al., 2023)

In addition to lactose hydrolysis,lactase can catalyze transgalactosylation reactions, in which lactose molecules are converted into galacto-oligosaccharides (GOS). These compounds function as prebiotics by selectively stimulating the growth of beneficial intestinal microorganisms such as *Bifidobacteria* and *Lactobacillus*, thereby supporting gut health and improving digestive balance.These enzymatic applications have significantly contributed to the expansion of lactose-free and functional dairy products, including lactose-free milk, yoghurt, and fermented dairy beverages, without compromising sensory quality orconsumer acceptance.

Figure 1: Lactose-free dairy products

Texture Improvement Through Enzymatic Modification

Texture is a critical determinant of consumer acceptance, and enzymes play a major role in modifying the structure and functionality of dairy products. Transglutaminase, an enzyme derived from microbial, plant, or animal sources, catalyzes the formation of covalent cross-links between milk proteins, thereby strengthening protein networks and improving product texture. In dairy processing, transglutaminase enhances gel firmness, viscosity, and water-holding capacity. Its application in Yogurt reduces syneresis and improves creaminess, while in Cheese it enhances curd strength, elasticity, and sliceability. Chymosin and proteases also influence texture by regulating protein breakdown during coagulation and ripening, contributing to desirable consistency and mouthfeel in different cheese varieties.

Nutritional Enhancement and Fat Modification

Enzyme Technology contributes significantly to the nutritional improvement of dairy products. During enzymatic hydrolysis, proteolytic enzymes break down milk proteins into smaller peptides known as bioactive peptides. These peptides exhibit several health-promoting properties, including antihypertensive effects by inhibiting angiotensin-converting enzyme (ACE), antioxidant activity through the neutralization of free radicals, and immunomodulatory functions that support immune responses. As a result, enzyme-treated dairy products can be developed as functional foods with enhanced health benefits.Lipase-catalyzed fat modification enables the production of structured lipids, which are specially designed fats in which fatty acids are rearranged to improve digestibility, nutritional value, and functional properties. These structured lipids can be enriched with omega-3 and medium-chain fatty acids,making them beneficial for specialized applications such as infant nutrition, medical foods,and dietary supplements. In addition, enzymatic fat modification improves lipid absorption and energy utilization while maintaining desirable sensory characteristics in dairy products.

Challenges and Future Prospects

Despite its numerous advantages, enzyme technology faces several challenges to enzyme stability, production cost, regulatory approval, and consumer perception. Some enzymes may lose activity under industrial processing conditions such as high temperature or pH variations, while large-scale enzyme production and purification can increase manufacturing costs. Advances in recombinant enzyme production, enzyme immobilization techniques, and process optimization are helping to overcome these limitations by improving enzyme stability, reusability, and processing efficiency. In addition, enzyme applications contribute to sustainability in dairy processing by reducing energy consumption, minimizing processing time, improving raw material utilization, and decreasing product waste. For example, enzymatic coagulation and accelerated ripening can shorten cheese manufacturing time, while lactose hydrolysis improves the utilization of dairy streams that would otherwise have limited consumer acceptance. Future developments are expected to integrate enzyme technology with advanced processing methods to further enhance sustainability, efficiency, and functional innovation in dairy products.

Enzyme technology has transformed dairy processing by enabling precise control over product quality, functionality, and nutrition. From cheese manufacturing and lactose-free milk to functional and health-oriented dairy products, enzymes provide versatile tools for value addition. As consumer demand for innovative, nutritious, and environmentally sustainable foods continues to grow, enzyme technology will remain central to the future of the dairy industry.

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FT 22

EDIBLE MICROLASERS FOR SMART FOOD TRACEABILITY AND QUALITY MONITORING

Introduction

Food traceability and authentication have become important concerns in modern food systems due to the globalized supply chains and increasing food frauds. Conventional labelling methods such as printed barcodes and QR codes are external to food products and also, they can be damaged, removed or even manipulated during the transportation and storage. Therefore, recent advances in food science and photonic technology have introduced edible microlasers as an innovative solution for secure identification and quality monitoring.

These microscopic laser devices are made up of using food-grade and biodegradable materials and it is allowing information to be embedded directly into food products without affecting their sensory or nutritional properties. In addition to identification, edible microlasers can also be function as internal food quality sensors and it has the potential for new opportunities for intelligent food systems.

What are Edible Microlasers?

Edible microlasers are microscopic optical devices produced entirely from edible and biodegradable materials. These systems are fabricated using natural substances such as olive oil droplets, gelatin, starch-based materials or edible polymers. Naturally occurring fluorescent compounds including chlorophyll and riboflavin (vitamin B_2) are often used as gain media to generate optical emissions.

The microlasers operate through optical cavity structures such as whispering gallery mode (WGM) resonators and Fabry-Perot cavities. When stimulated using an external light source, they emit highly specific optical signals or laser spectra. These optical signatures can later be detected and decoded using optical scanners or spectroscopic instruments.

Edible microlasers are extremely small that often smaller than a grain of sand. It is allowing them to be incorporated into food such as dairy products, beverages, fruit preparations and semi solid food without altering their appearance, taste or texture.

Edible Barcodes and Internal Food Identification

One of the most promising applications of edible microlasers is their use as edible barcodes for internal food identification. Unlike conventional surface la-

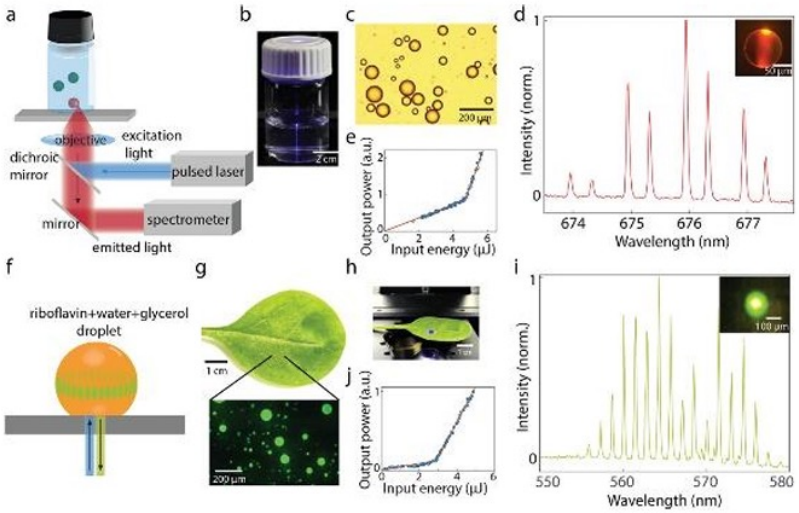

Figure 1: Experimental setup and lasing behaviour of edible WGM microlasers based on dye-doped droplets, demonstrating optical emission, threshold characteristics and sensing capability. Adapted from Anwar et al. (2025).

bels, edible barcodes are embedded directly into the food matrix. This makes them resistant to tampering, environmental damage and counterfeiting.

These edible optical barcodes can encode valuable information related to the food product, including the production date, batch number, manufacturer details, expiry date and storage history. During different stages of the supply chain, the encoded optical signs can be read using spectroscopic techniques or portable optical scanners.

Several sectors in the food industry could benefit from this technology. In dairy products, edible microlasers could help identify product origin and storage conditions. In beverage manufacturing, internal optical codes could improve batch tracking and reduce product substitution. Similarly, high-value products such as organic foods, nutraceuticals and premium fruit products could use edible barcodes to prevent food fraud and authenticate product quality.

Recent studies have demonstrated that edible microlasers can maintain stable optical properties even after prolonged storage at room temperature. According to Anwar et al. (2024), edible laser systems fabricated from natural materials showed successful optical decoding while maintaining food compatibility.

Sensing Applications in Food Quality Monitoring

Edible microlasers can also function as highly sensitive food quality sensors. Because of their narrow optical emission properties, small environmental changes within the food matrix can produce measurable shifts in laser signals. This enables real-time monitoring of food quality and storage conditions.

Researchers have investigated edible microlasers for monitoring several important food quality parameters including sugar concentration, pH changes, temperature fluctuations and microbial activity. For example, changes in sugar concentration during fermentation processes can alter the refractive index of the surrounding environment and it results in detectable optical shifts. Similarly, pH-responsive edible materials have shown potential for detecting spoilage in dairy products and fermented foods.

Temperature-sensitive edible microlasers can also be used to monitor cold chain integrity during the food transportation and storage. In some studies, irreversible changes in optical signals were observed when food products experienced temperature abuse. It allows rapid identification of cold chain failure. Such applications could significantly improve food safety and reduce economic losses

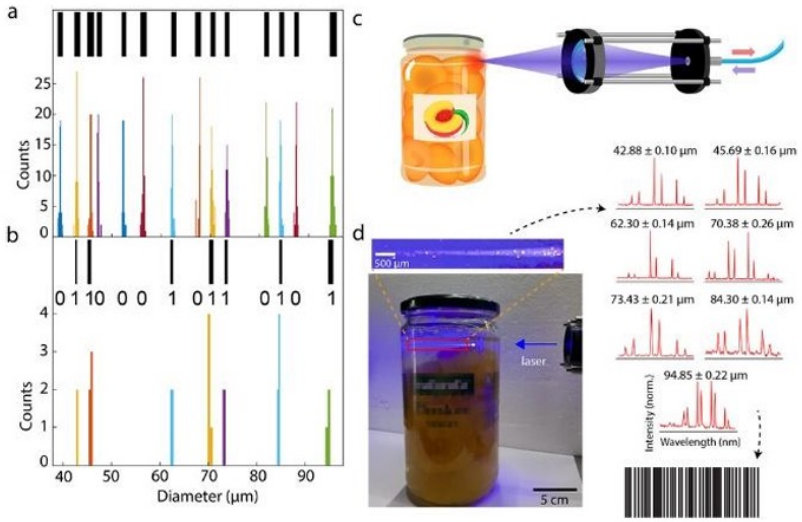

Figure 2: Optical encoding and decoding of edible barcodes using chlorophyll-doped oil droplets. The system demonstrates droplet-size-based information storage, optical fiber readout, and successful reconstruction of encoded messages in a food matrix. Adapted from Anwar et al. (2025).

associated with spoilage.

Advantages of Edible Microlasers

Edible microlasers offer many important advantages for modern food systems. One important advantage is their high sensitivity to environmental changes. Even small variations in temperature, pH or chemical composition can produce detectable optical responses and it enables accurate monitoring of food quality.

Another important advantage is their food-safe and biodegradable nature. Since these systems are made up of using edible materials and naturally occurring compounds that they can be safely incorporated into food products without making additional packaging waste. This supports sustainability goals and environmentally friendly food technologies.

Edible microlasers are also invisible to consumers and do not significantly affect the taste, texture or appearance of the food products. Their placement within the food matrix beneficial for improving security because the embedded optical signatures are difficult to counterfeit. Premium food products such as organic foods, nutraceuticals and pharmaceutical foods could especially benefit from this feature. Furthermore, early detection of spoilage and storage failures could help reduce food waste and improve supply chain efficiency.

Challenges and Limitations

Despite the high potential edible microlasers have several challenges when it comes to large scale commercial implementation. Current fabrication methods are still mainly limited to laboratory scale research and require further optimization for commercial food processing environments.

Another challenge is the complexity of optical reading systems. Even though spectroscopic techniques can successfully decode optical signatures, simpler and more cost-effective portable scanners are required for practical industrial applications.

Regulatory approval also remains an important consideration. Since edible microlasers are incorporated directly into foods, regulatory authorities must carefully evaluate their long-term safety, toxicity and stability before commercialization.

Consumer acceptance may also create challenges. Some consumers may feel uncomfortable about the idea of microscopic laser systems being embedded inside food products even though the materials are completely edible and safe. Concerns related to misunderstanding of the term "laser" may reduce public acceptance. Therefore, proper consumer education and awareness programs will be important for improving confidence within them.

Future Prospects

Edible microlasers represent a shift toward packaging-free smart food monitoring. With more development, they could be utilized in reducing food fraud, improving supply chain transparency and minimizing global food waste. Future research may focus on developing more affordable fabrication methods, portable optical detection systems and multifunctional edible sensors capable of monitoring several quality parameters simultaneously.

Conclusion

Edible microlasers are an emerging technology in food science and technology. By embedding identification and sensing systems directly into food using safe, digestible materials, they offer a secure, sustainable and innovative solution for food traceability and safety. Although challenges related to commercialization, regulation and consumer acceptance still exist. Continued research and technological refinement may soon make edible photonic systems a standard tool in next generation food systems.

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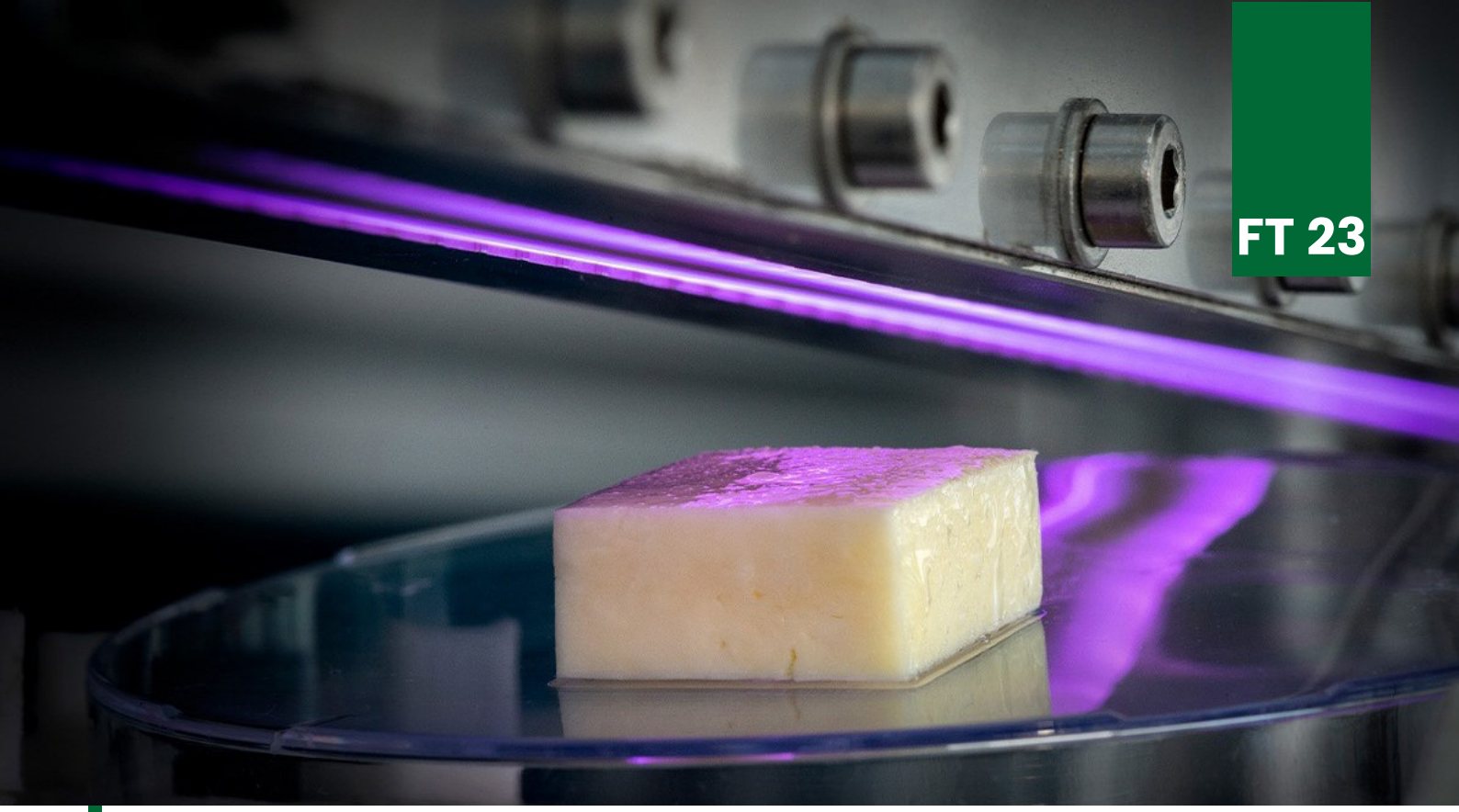


BEYOND HEAT

HOW NON-THERMAL TECHNOLOGIES ARE TRANSFORMING FOOD PRESERVATION.

Introduction:

Non-thermal food preservation technologies have emerged as innovative alternatives to conventional heat-based processing methods such as pasteurization, sterilization, and canning. For many years, thermal processing has played a critical role in ensuring microbial safety and extending the shelf life of foods. However, exposure to high temperatures often results in undesirable changes, including nutrient degradation, texture softening, color deterioration, and loss of natural flavor compounds (Aadil et al., 2013). In modern food markets, consumers increasingly prefer minimally processed foods that retain their fresh-like sensory and nutritional qualities while remaining safe for consumption. This growing demand has accelerated research and industrial interest in non-thermal food processing technologies.

Non-thermal food processing involves the inactivation of microorganisms and enzymes using physical mechanisms other than heat. Unlike conventional thermal treatments, these technologies are designed to preserve sensitive bioactive compounds, vitamins, pigments, and volatile flavor molecules while maintaining food safety and quality (Barba et al., 2017). Different non-thermal methods operate through distinct mechanisms and are applied according to the characteristics of specific food products.

High Pressure Processing (HPP), for example, uses extremely high hydrostatic pressure to disrupt microbial cell membranes and enzyme activity without significantly affecting the covalent bonds of nutrients and flavor compounds. Pulsed Electric Field (PEF) technology works by applying short bursts of high-voltage electric pulses that disrupt microbial cell membranes through a process called electroporation.

Cold plasma technology has also gained attention as an emerging non-thermal approach for surface decontamination of fresh produce, spices, and packaging materials. Reactive ions and free radicals generated by plasma interact with microbial cells and lead to their destruction while operating at near-room temperatures (Misra et al., 2011). In addition, ultrasound processing utilizes acoustic cavitation produced by high-frequency sound waves to improve extraction efficiency, emulsification, drying, and microbial reduction in food systems (Chemat et al., 2011).

Due to their ability to ensure safety while preserving nutritional and sensory quality, non-thermal food processing technologies are increasingly recognized as sustainable and consumer-oriented solutions for the future food industry.

High Pressure Processing (HPP):

One of the most commercially successful non-thermal technologies is high pressure processing (HPP). This method exposes food to extremely high hydrostatic pressures, typically between 100-800 MPa. Such pressure disrupts the cellular structure of microorganisms like vegetative bacteria, spoilage organisms, pathogens, some viruses and fungi without affecting covalent bonds in nutrients. Therefore, HPP has strong potential for improved organoleptic properties and reduced loss of nutrients and flavors, while achieving extended shelf life and improved safety. HPP highly suitable for products such as fruit juices, ready-to-eat meats, seafood, and guacamole, where fresh taste and texture are important quality attributes (Considine et al., 2008).

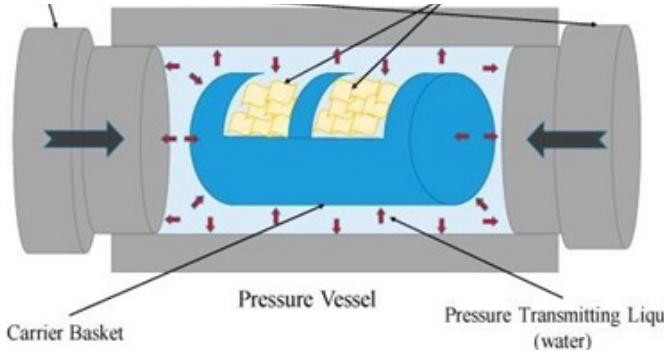


Figure 01: High Pressure Processing Machine

Pulsed Electric Field (PEF):

Pulsed Electric Field (PEF) is another innovative mild food preservation technology, which applies very short pulses of high-voltage direct electric current that cause electroporation and damage microbial cell membranes, making it particularly effective for liquid and semi-liquid foods such as fruit juices, milk, and liquid eggs (Toepfl et al., 2006). PEF has the potential to be used as a replacement for thermal pasteurization for the treatment of either acidic foods or foods which are subsequently stored at refrigerated temperatures. Beside microbial inactivation, PEF can enhance juice extraction efficiency and improve mass transfer process making it attractive for industrial applications.

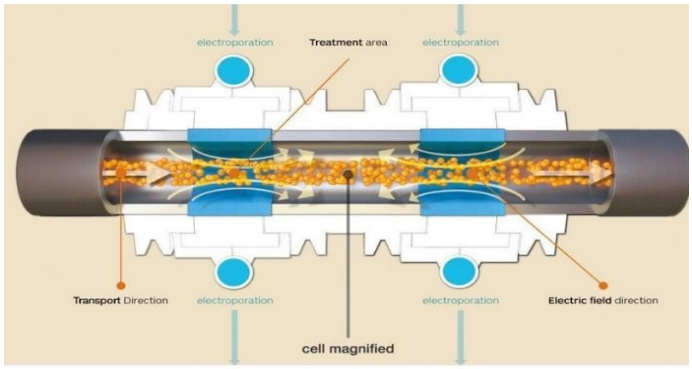


Figure 02: Pulsed Electric Field System

Cold Plasma Technology:

Cold plasma technology is a newer and rapidly emerging non-thermal method. Plasma is a gaseous mixture that is particularly or completely ionized. Supplying energy input, solids change into liquids, which subsequently turn into gases. Plasma consists of activated molecules, free radicals, positive ions, negative ions, and neutral atoms which effectively inactivate microorganisms in the food surface. Many of gases like O2, H2, He, Ne, CH4, and NH3 can be converted into plasma. Cold plasma is especially useful for surface decontamination of fresh produce, spices and packing materials. Although still largely at the research and pilot stage, its potential for food safety improvement is significant.

Ultrasound Processing:

Ultrasound processing uses high frequency sound waves in the range of 20-100 kHz to create cavitation bubbles in liquids. When these bubbles collapse, they generate localized pressure and temperature changes that can disrupt microbial cells and enhance physical processes. The inactivation of microorganisms in liquid foods and reduction of heat requirement for microbial inactivation are the potential benefits of ultrasound processing. This technology is commonly applied in extraction of bioactive compounds, emulsification and improving drying and freezing efficiency.

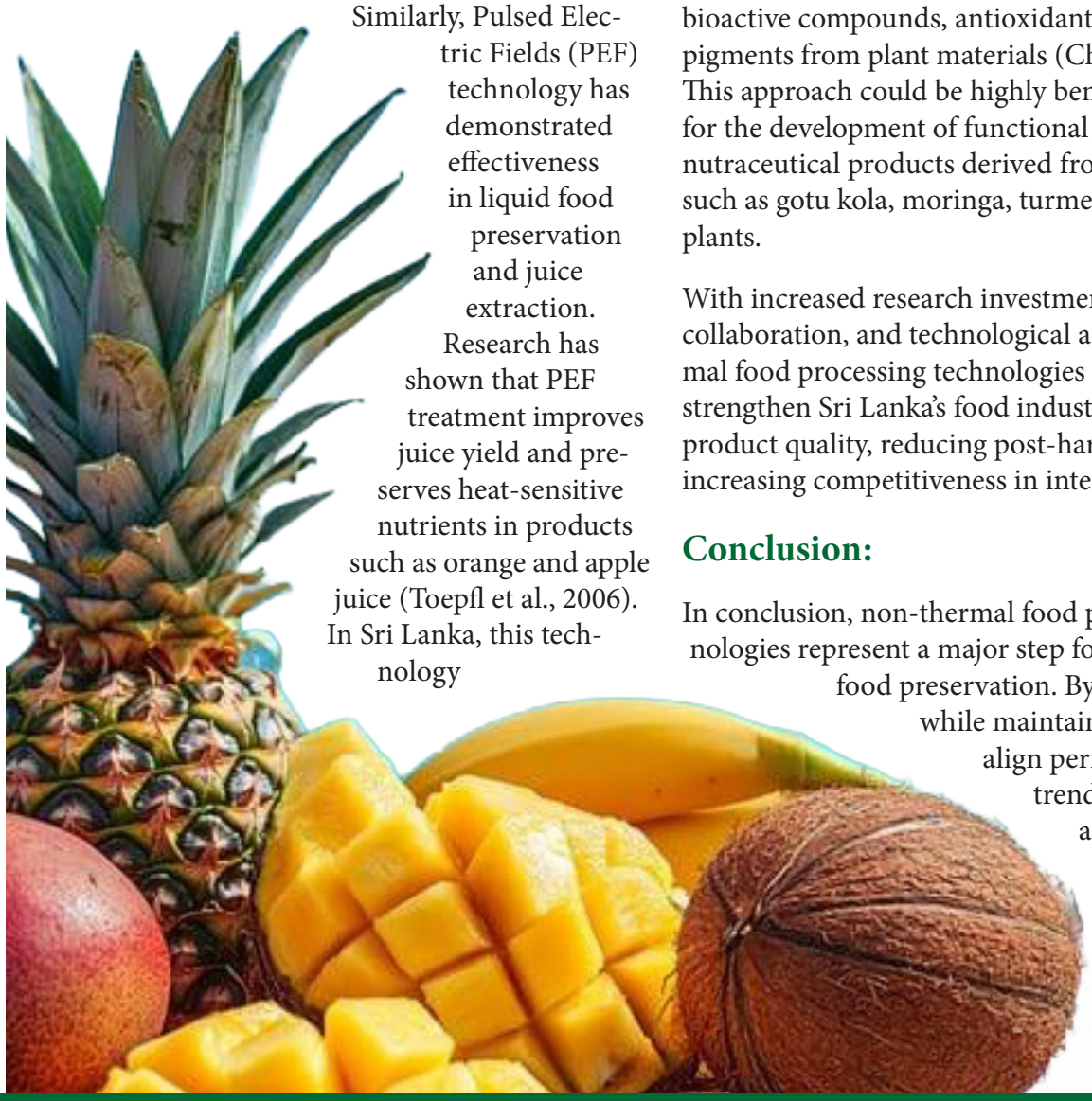
Benefits and Challenges:

Compared to traditional thermal processing, non-thermal technologies offer several advantages. They retain heat-liable nutrients such as vitamins and antioxidants, preserve natural texture and color, and reduce the need for chemical preservatives, more energy-efficient and environmentally friendly. However, these technologies also face challenges. They include high equipment cost, limited industrial expertise and

regulatory barriers, particularly in developing countries, and are not suitable for all food types; careful process optimization is required.

Future Potential and Sri Lankan Context:

In the Sri Lankan context, non-thermal food processing technologies hold considerable potential for improving the quality, safety, and export value of locally produced foods. Sri Lanka is rich in tropical fruits such as mango, pineapple, papaya, wood apple, and king coconut, many of which experience significant post-harvest losses due to rapid spoilage and inadequate preservation methods. High Pressure Processing (HPP) could be effectively applied in the production of fresh fruit juices and ready-to-eat products, as studies have shown that HPP-treated fruit juices retain their natural color, vitamin content, and fresh flavor while achieving extended shelf life and microbial safety (Considine et al., 2008). This technology would be particularly valuable for Sri Lanka's growing fruit beverage and export industries.



Similarly, Pulsed Electric Fields (PEF) technology has demonstrated effectiveness in liquid food preservation and juice extraction. Research has shown that PEF treatment improves juice yield and preserves heat-sensitive nutrients in products such as orange and apple juice (Toepfl et al., 2006). In Sri Lanka, this technology could be applied to value-added beverages produced from tropical fruits and herbal drinks, helping manufacturers maintain fresh-like quality while reducing thermal damage.

Cold plasma technology also presents promising opportunities for Sri Lankan food industries, especially in the decontamination of spices, fresh-cut fruits, and packaging materials. Sri Lanka is internationally recognized for spice exports such as cinnamon, pepper, and cloves, where microbial contamination can reduce export quality. Studies have reported that cold plasma treatment effectively reduces microbial populations on spice surfaces without significantly affecting aroma and bioactive compounds (Misra et al., 2011). Therefore, the technology could support safer and higher-quality spice exports while minimizing reliance on chemical fumigation.

In addition, ultrasound processing has gained attention for enhancing extraction efficiency and improving food processing operations. Ultrasound-assisted extraction has been successfully applied to recover bioactive compounds, antioxidants, and natural pigments from plant materials (Chemat et al., 2011). This approach could be highly beneficial in Sri Lanka for the development of functional beverages and nutraceutical products derived from local resources such as gotu kola, moringa, turmeric, and herbal plants.

With increased research investment, industrial collaboration, and technological adoption, non-thermal food processing technologies could significantly strengthen Sri Lanka's food industry by improving product quality, reducing post-harvest losses, and increasing competitiveness in international markets.

Conclusion:

In conclusion, non-thermal food processing technologies represent a major step forward in modern food preservation. By ensuring safety while maintaining quality, they align perfectly with future trends in sustainable and consumer-focused food production.

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SAFE ON PAPER, RISKY IN REALITY

THE BLIND SPOTS IN TODAY'S FOOD SAFETY SYSTEMS

Food safety regulations have protected public health for decades, reducing foodborne illness and improving hygiene across supply chains. On paper, today's systems appear comprehensive. But in reality, rapid changes in how food is produced, processed, and delivered have exposed serious blind spots. Several major food safety incidents in recent years show what happens when standards fail to keep pace with modern risks.

Old Rules in a Digital Food World

Food laws in many countries were designed for traditional restaurants, markets, and factories -no**t cloud kitchens, home-based food sellers, or app-based delivery systems**. These newer food business models may operate with minimal physical infrastructure and inconsistent inspection coverage.

Without updated licensing systems and digital monitoring, some food businesses remain largely invisible to regulators, increasing the risk of poor hygiene, temperature abuse, and unsafe handling practices.

Chemical and Emerging Hazards: A Growing Concern

Food safety systems have historically focused on microbial contamination, but chemical hazards often receive less attention. Pesticide residues, packaging migrants, processing contaminants, and environmental pollutants can enter the food chain and pose long-term health risks.

A broader issue is that many chemicals used in food production and packaging enter the market without comprehensive modern safety reassessments. Unlike food poisoning, chemical exposure can lead to chronic diseases that develop slowly, making these risks harder to detect and regulate effectively.

Enforcement Gaps: When Standards Exist but Protection Fails

Strong laws alone do not guarantee safety; enforcement capacity is equally vital. **The 2017–2018 listeriosis outbreak** in South Africa, which caused over 200 deaths, was traced to contaminated processed meat from a facility where environmental pathogens went undetected. This tragedy exposed critical gaps

in routine monitoring and traceability.

Similarly, in 2021, the Sri Lanka Standards Institution (SLSI) and the Customs Department detected Aflatoxin B1 and B2 levels far exceeding permissible limits in several imported oil containers, sparking a national crisis over testing protocols.

In 2022, contaminated infant formula in the United States led to fatal Cronobacter infections, revealing sanitation failures even within advanced regulatory systems. These events demonstrate that inspection frequency, laboratory capacity, and proactive monitoring are just as essential as the written standards themselves.

Traceability That Stops at Paperwork

Modern food supply chains are global and complex, yet traceability in many systems still depends heavily on manual documentation.

In 2022, a multi-country *Salmonella* outbreak linked to **chocolate products in Europe** spread across several countries before being fully controlled. The delay in stopping contaminated products from reaching consumers highlighted the importance of faster, digital traceability and quicker corrective action when contamination is detected internally.

Without integrated digital systems to track batches, temperatures, and distribution routes, recalls become slower and more consumers are exposed before action is taken.

Audit Culture vs. Food Safety Culture

Many food businesses prepare for inspections by focusing on documentation rather than real risk reduction. This creates a **"tick-the-box" culture,** where passing audits becomes more important than preventing hazards.

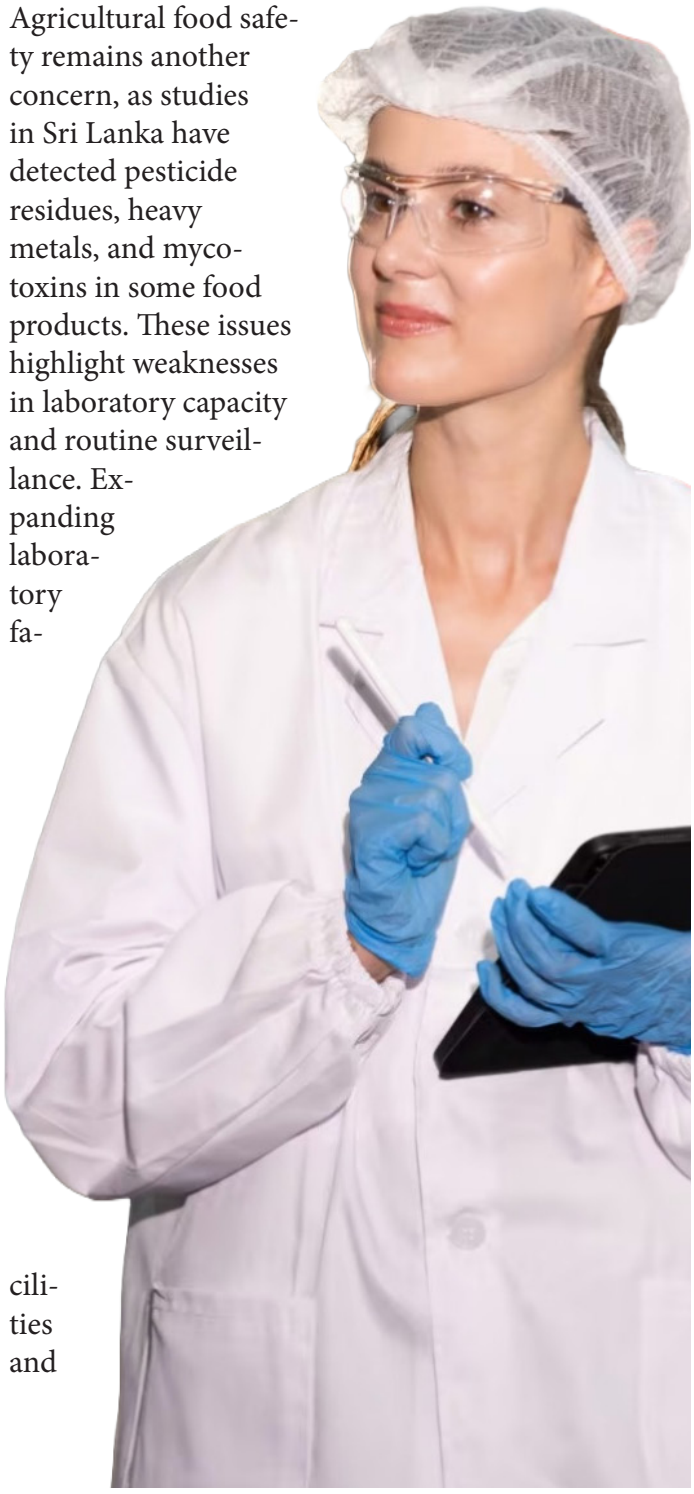
True food safety depends on leadership commitment, staff training, and a workplace culture where risks are identified and corrected early. Standards rarely measure this food safety culture, even though it strongly influences real-world outcomes.

Sri Lankan Context: Local Challenges in a Global Pattern

Sri Lanka reflects many global food safety challenges, with several important gaps still remaining in the system. Although the country operates under the Food Act of 1980, the law does not fully address modern food systems, online food businesses, or emerging contaminants. Updating regulations and introducing better monitoring and traceability systems are therefore essential.

Food safety enforcement is also limited by shortages of trained Public Health Inspectors and resources, making regular inspections difficult, especially for informal vendors and small eateries. Increasing trained personnel and strengthening inspection facilities would improve enforcement.

Agricultural food safety remains another concern, as studies in Sri Lanka have detected pesticide residues, heavy metals, and mycotoxins in some food products. These issues highlight weaknesses in laboratory capacity and routine surveillance. Expanding laboratory facilities and



continuous monitoring programs would help control such risks more effectively.

Moving Toward Food Safety That Works in Reality

To close the gap between "safe on paper" and "safe in practice," food safety systems must evolve. Priorities include:

- Updating laws to cover digital and informal food sectors
- Strengthening inspection services and laboratory infrastructure
- Improving chemical risk monitoring & digital traceability systems
- Promoting food safety culture, not just audit compliance

Food safety can no longer rely only on traditional approaches. In a rapidly changing food world, protection must be proactive, science-based, and technology-driven. Only then can standards truly deliver what they promise: safe food for everyone.

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SAFE ON PAPER, RISKY IN REALITY

THE BLIND SPOTS IN TODAY'S FOOD SAFETY SYSTEMS

In the modern food industry, plant-based milk alternative beverages are the most demanded products compared to animal products due to increasing lactose intolerance, sustainability issues, and ethical concerns. Due to allergenicity, environmental burden, and nutritional limitations of such soy, almond, and oat milk, affect to the dairy industry. The purpose of this article was to assess the suitability of yam milk as a sustainable dairy alternative when compared to existing plant-based milk drinks like soy, almond, and oat milk. The consumption of plant-based milk alternatives has become prevalent due to the growing number of lactose-intolerant individuals, as well as for several other reasons. Yams are members of the *Dioscoreaceae* family and belong to the genus Dioscorea. They are known as a functional tuber and have the potential for the development of a yam plant-based milk beverage.

Yams are also an abundant source of starch, dietary fiber, resistant starch, and bioactive substances like diosgenin, phenolic, and flavonoid compounds, which have antioxidant and beneficial effects on human health (Wang et al., 2025). Freshly harvested yam tubers are good sources of starch, dietary fiber, and other bioactive components. But when processing yam milk products, processes such as blanching, homogenization, filtering, and pasteurizing are done to increase product quality. Fermentation enhances the sensory characteristics, nutritional content, texture, and microbiological quality of plant-based beverages (Tangyu et al., 2019).

On the agricultural side, yams are a resilient crop that grows in tropical environments, and they are commonly grown in countries such as Sri Lanka. In addition to their nutritional value, yams remain maximized in value-added food applications and are still consumed in traditional forms. Taking into consideration their application in plant-based dairy production can enhance the value of the crop and promote sustainable food innovation. Starch based plant beverages may gelatinize during heating as starch molecules absorb water and swell, which in turn increases viscosity. Retrogradation may be achieved during storage by the association of starch molecules, leading to sedimentation.

Several techniques have been employed to increase the physical stability of plant-based milk drinks and minimize phase separation, including homogenization, emulsification, and stabilization with stabilizers.

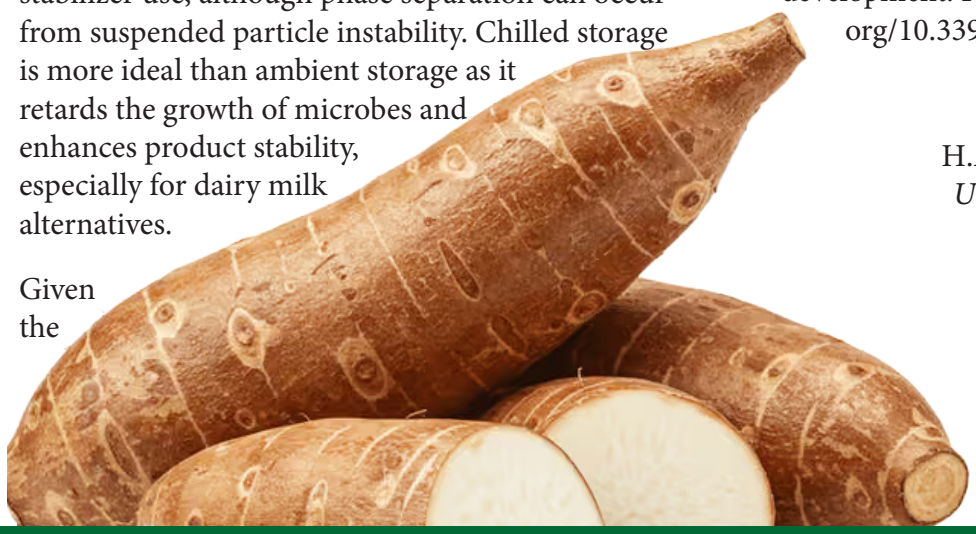
Homogenization increases the stability of the emulsion and the mouthfeel of plant based milk drinks, which is achieved through the reduction of particle size; in addition, fermentation by microorganisms such as lactobacilli can increase flavor, texture, nutrient profile, and safety.

While plant-derived milk substitutes contain dietary fibers and bioactive compounds, most commercial brands are not nutritionally comparable to dairy milk because they lack the required protein quality, vitamin B12, vitamin D, and calcium (Tangyu et al., 2019).

Many ways can be applied to overcome problems with processing and stability in yam milk products. For instance, the application of stabilizers, including xanthan gum, carrageenan, and guar gum, will promote emulsion stability and prevent phase separation. Moreover, nutritional enhancement through protein enrichment, addition of calcium, vitamin B12, and vitamin D can help to improve the nutritional value of yam milk, which is comparable to dairy milk. The yam root milk may have some earthy flavors and textural problems that will reduce the level of acceptance by the consumers.

There are many ways to enhance the flavor of the drink, such as vanilla, cocoa, cinnamon, or fruit flavors. Homogenization, stabilizers, and balancing the ratio between solids and water can help to improve the texture. Acceptance tests also have to be done to determine the preference for taste, smell, color, and texture. Plant-based milk alternatives are highly susceptible to spoilage by microorganisms because of their high moisture and nutrient content. Pasteurization, hygienic processing, and suitable packaging are essential for safety and shelf-life. Emulsion stability can be improved by homogenization and stabilizer use, although phase separation can occur from suspended particle instability. Chilled storage is more ideal than ambient storage as it retards the growth of microbes and enhances product stability, especially for dairy milk alternatives.

Given the cultivation of yams, yam-based plant milk could be identified as a product for the Sri Lankan market, given the demand for plant-based milk products. The product would be enhanced by considering developments in process, fermentation, and fortification techniques. Yam-based plant milk is an innovative, healthy, and sustainable product in comparison to dairy and other plant based milk products. Future research on yam-based milk should be centered on establishing the effectiveness of their nutritional and health value through clinical studies because of the bioactive components present in Dioscorea. Research will be conducted on food fortification, food processing techniques, shelf-life extension, and industrial manufacturing. Consumer testing on taste and texture improvement is crucial in the development and commercialization of yam-based dairy products.



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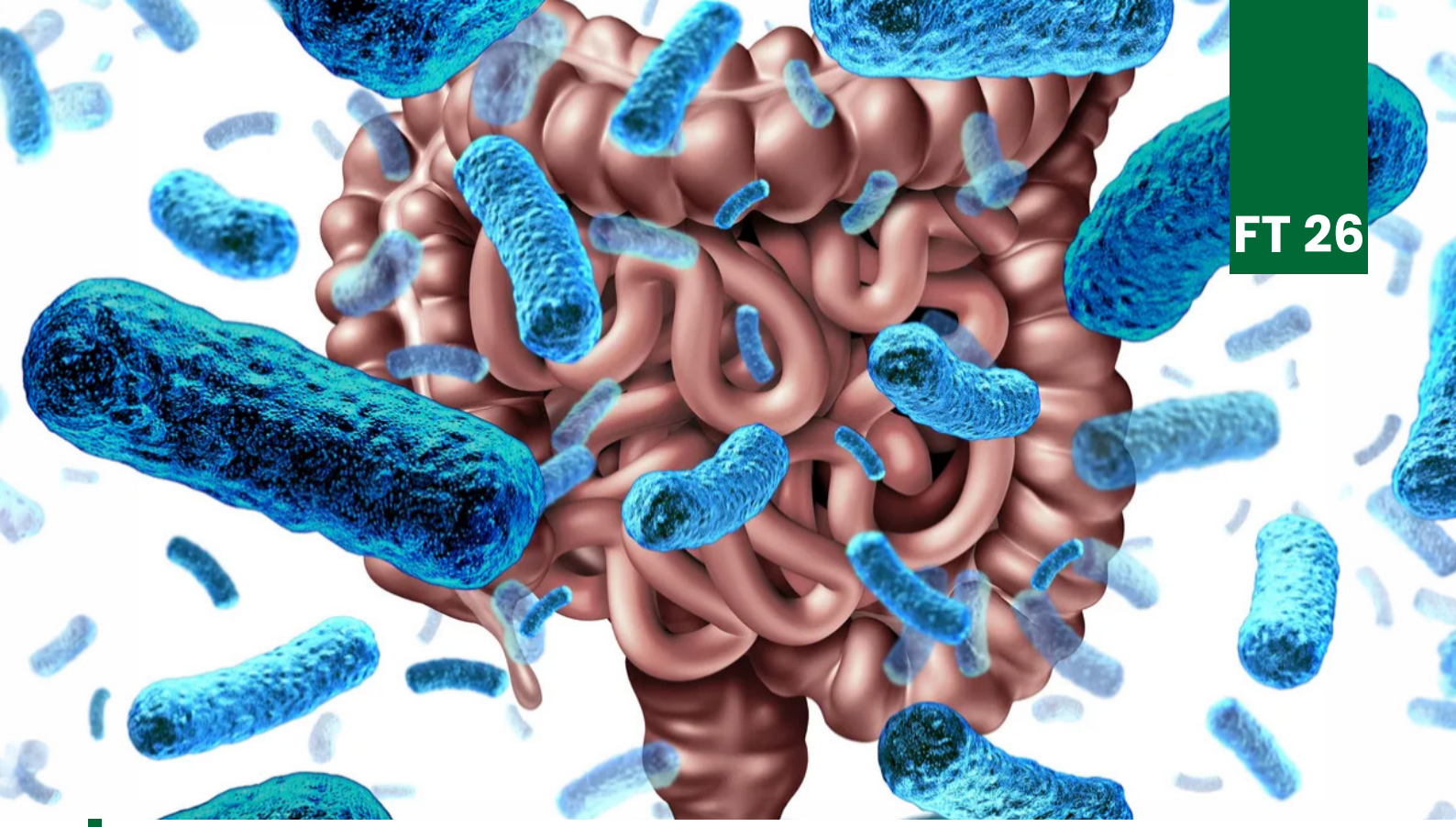


COULD YOUR GUT BE AFFECTING YOUR MOOD?

Feeling anxious? Struggling with depression? We often blame stress, lack of sleep, work pressure, or difficult life experiences. While these are all important factors, there may be another surprising contributor hiding deep inside your body.

What if trillions of tiny microorganisms living in your gut, collectively known as the gut microbiome, were also affecting how you think, feel, and respond to stress?

The gut microbiome continuously communicates with the brain through the 'gut-brain axis'. Growing research suggests that when this microbial community is out of balance, it may contribute to anxiety, depression, and other mental health conditions.

Could improving your gut health also help improve your mental well-being? Let's understand the interesting science behind this hidden connection.

The Gut and Brain Are Constantly Talking

The gut–brain axis is a two-way communication system that involves the nervous system, hormones, the immune system, and countless chemical messengers. One of the most important communication pathways is the 'vagus nerve', often called the body's "information superhighway." It carries signals between the digestive tract and the brain, allowing each organ to influence the other.

Gut microorganisms also produce a variety of biologically active compounds. Some bacteria generate short-chain fatty acids such as butyrate, acetate, and propionate, which help maintain intestinal health while reducing inflammation. Others influence the production of neurotransmitters, including serotonin, dopamine, and gamma-aminobutyric acid (GABA), that play essential roles in regulating mood, sleep, motivation, and emotional well-being. Interestingly, nearly 90% of the body's serotonin, a neurotransmitter often associated with happiness, is produced in the gastrointestinal tract (Hwang & Oh, 2025).

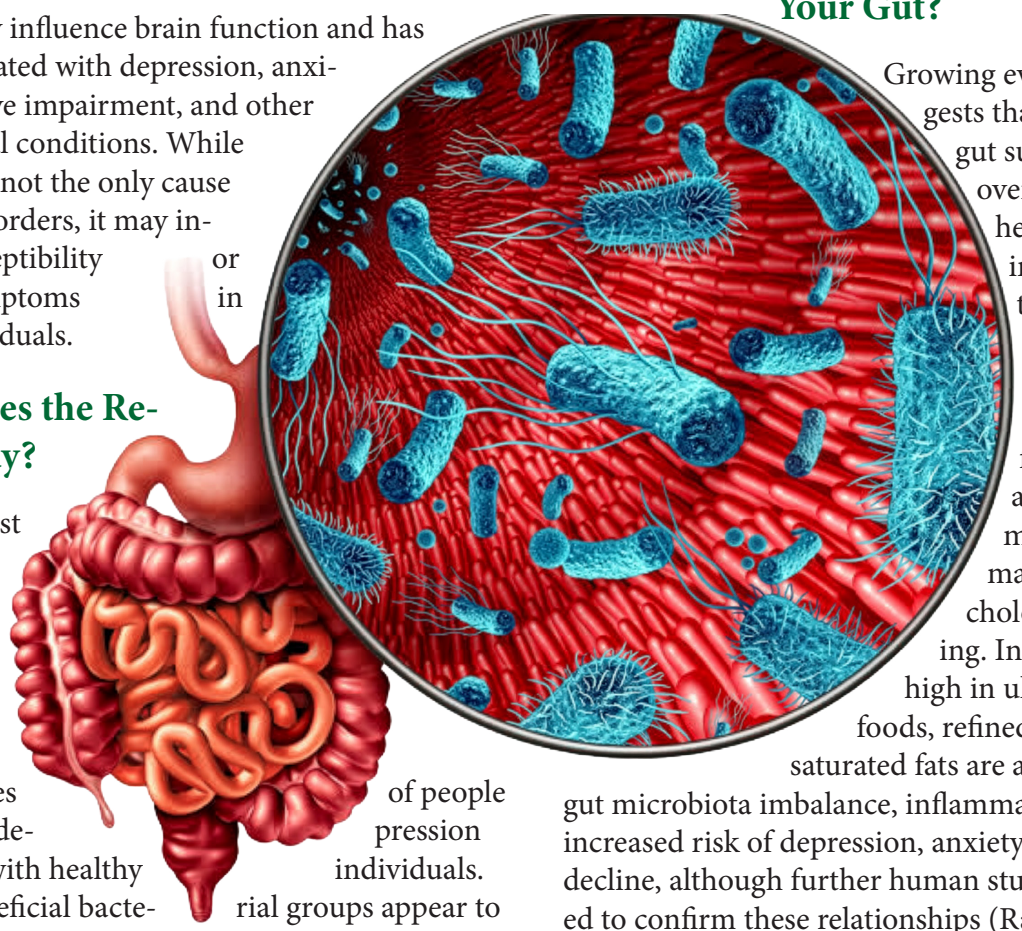
When the Microbial Balance Is Disturbed

A healthy gut microbiome resembles a diverse and balanced ecosystem. However, factors such as poor diet, chronic stress, frequent antibiotic use, inad-

equate sleep, excessive alcohol consumption, infections, and a sedentary lifestyle can disrupt this balance-a condition known as 'dysbiosis'.

When dysbiosis occurs, beneficial bacteria decrease while potentially harmful microorganisms increase. This imbalance may weaken the intestinal barrier, allowing bacterial products to enter the bloodstream and trigger low-grade inflammation throughout the body.

Researchers have discovered that chronic inflammation may influence brain function and has been associated with depression, anxiety, cognitive impairment, and other neurological conditions. While dysbiosis is not the only cause of these disorders, it may increase susceptibility or worsen symptoms in some individuals.



What Does the Research Say?

Over the past decade, scientists have found marked differences in the gut microbiomes of people living with depression compared with healthy individuals. Several beneficial bacterial groups appear to be reduced, while others become more abundant.

Animal studies provide even more compelling evidence. In one remarkable experiment, researchers transferred gut microbes from individuals with depression into germ-free laboratory mice. Surprisingly, the mice show behaviors resembling depression, suggesting that gut microorganisms can influence behavior under certain conditions (Zheng et al., 2016).

Human studies have also shown promising results. Some clinical trials indicate that consuming specific probiotic strains may reduce stress, improve mood, and lessen anxiety in certain people. These beneficial microorganisms are sometimes referred to as 'psychobiotics' because of their potential influence on mental health (Mosquera et al., 2024).

However, scientists caution that this field is still evolving. Depression and anxiety are highly complex conditions influenced by genetics, life experiences, hormones, physical health, social circumstances, and many other factors. The gut microbiome is one important piece of a much larger puzzle-not a standalone explanation.

Can You Feed Your Mind by Feeding Your Gut?

Growing evidence suggests that a healthy gut supports both overall and brain health. Diets rich in fruits, vegetables, whole grains, legumes, nuts, seeds, and fermented foods promote a diverse gut microbiome and may improve psychological well-being. In contrast, diets high in ultra-processed foods, refined sugars, and saturated fats are associated with gut microbiota imbalance, inflammation, and an increased risk of depression, anxiety, and cognitive decline, although further human studies are needed to confirm these relationships (Randeni & Xu, 2025).

A New Way of Thinking About Mental Health

For centuries, medicine treated the brain and the digestive system as separate organs with separate functions. Modern science is revealing a far more interconnected picture.

Your gut is not simply where food is digested. It is a dynamic ecosystem that communicates with your immune system, endocrine system, and nervous system. The trillions of microbes living within you actively participate in processes that influence your health.

This does not mean that depression or anxiety can be cured by eating yogurt or taking a probiotic supplement. Mental health disorders require proper evaluation and, when necessary, evidence-based treatment provided by qualified healthcare professionals. However, maintaining a healthy gut microbiome may become an important component of a comprehensive approach to supporting mental well-being.

The next time you think about mental health, remember that the conversation may not begin in your head alone. Sometimes, it starts much deeper-within the microscopic world living inside your gut.

References

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